

**Fakultät für Physik und Astronomie
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**Deep Learning in the Depths of Space:
U-Net-Based Denoising for Shallow Extragalactic Surveys based on
HST Infrared Images**

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Zusammenfassung

In dieser Arbeit wird die Anwendung eines U-Net-basierten *Convolutional Neural Network* (CNN) zur Rauschminderung astronomischer Bilder untersucht, wobei der Schwerpunkt auf extragalaktischen *Surveys* mit Infrarotdaten des Hubble Space Telescope (HST) F160W-Filters liegt. Das Netzwerk wurde auf einem Datensatz von 1.136 HST-Bildern trainiert, die durch sukzessive Hinzufügung von Gaußschem Rauschen künstlich degradiert wurden, um die Rauschdynamik bei reduzierten Belichtungszeiten zu simulieren – bis zu einem Faktor von 40 unterhalb der ursprünglichen Belichtungszeit. Die Ergebnisse zeigen, dass das Modell für Objekte mit einer AB-Magnitude von 25 oder heller zuverlässige Rekonstruktionen liefert. Der Strukturelle Ähnlichkeitsindex (SSIM) erreicht dabei hohe Werte von 0,99. Zudem werden strukturelle Parameter wie der Kron-Radius mit einem mittleren Fehler von 0,41'' ohne systematische Verschiebung sicher geschätzt. Das Modell erzielt eine Trefferquote von 0,83, eine Präzision von 0,89 und ein F1-Maß von 0,86, was auf ein gutes Gleichgewicht zwischen der Erkennung echter Objekte und der Minimierung von Artefakten hinweist. Die Fluxes werden jedoch systematisch um 9 % überschätzt, mit einem höheren mittleren Fehler von 20 %, insbesondere bei schwachen Objekten. Die Leistung nimmt für Objekte mit AB-Magnitude > 25 aufgrund des erhöhten Rauschens und der Unsicherheit bei der Erkennung deutlich ab. Diese Einschränkungen sollten als Obergrenze für die derzeitigen Fähigkeiten des Netzwerks ohne weiterführende Optimierungen interpretiert werden.

Abstract

This thesis investigates the application of a U-Net-based convolutional neural network (CNN) architecture for denoising astronomical images, specifically focusing on extragalactic surveys using infrared data from the Hubble Space Telescope (HST) F160W filter. The network was trained on a dataset of 1,136 HST images, which were successively corrupted with Gaussian noise to simulate the noise dynamics at lower exposure times, up to 40 times lower than the original exposure. Results show that the model performs well for objects with an AB magnitude of 25 or brighter, achieving high Structural Similarity Index (SSIM) values (0.99) and precise Kron radius estimations (mean error of 0.41"). Furthermore, the model exhibited a precision of 0.83, recall of 0.89, and an F-measure of 0.86, indicating a good balance between detecting real objects and minimizing artifacts. However, fluxes were consistently overestimated by 9%, with a higher error margin of 20%, particularly for faint objects. The performance declines significantly for objects fainter than 25 AB magnitude due to increased noise and uncertainty in faint object detection. These limitations should be interpreted as an upper bound for the network's current capabilities without further significant modifications.

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Chapter 1

Introduction

The increase in computing power over the last decade has enabled scientists to couple human intellect with vast computational power paving the way for doing science in ways not possible before. From detecting early cancerous cells to helping us writing efficient code to enhancing image quality, artificial intelligence has become an indispensable part of our lives. It is also increasingly being adapted in the astronomical community. Some areas where artificial intelligence techniques are used in astronomy include automating data processing, enhancing image analysis, and accelerating discoveries in fields like galaxy classification, exoplanet detection, and gravitational wave identification. Models such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), analyze vast datasets from telescopes, can improve Signal-to-Noise-Ratio (SNR) of shallow images. Beside the U-Net architecture which will be used in this thesis, the use of machine and deep learning expands to different state-of-the art methods based on the needs of the study and the properties of the data.

Deep learning (DL) techniques are integrated into Python packages so as to allow scientists who have less experience with DL to use them through the Python API (Application Programming Interface). One such case is the **Zoobot** which is designed for astronomers to analyze galaxy images using DL and makes use of Bayesian CNNs which introduces Bayesian inference to the CNNs and thereby error estimations (Shridhar et al., 2019). By using a Bayesian CNN, Zoobot helps solve tasks like identifying merging galaxies or counting spiral arms by allowing users to finetune pretrained models. These finetuned models perform better and require fewer new labels than training from scratch. Zoobot’s models are pretrained on data from Galaxy Zoo volunteers, with up to 92 million responses (Walmley et al., 2023). Applying Zoobot to HST F814W images, (O’Ryan et al., 2023) made interaction predictions for 126 million sources in the Hubble Source Catalogue which lead to the identification of 21,926 interacting galaxy systems, most of which are at redshifts $z < 1$. Notably, 65% of these systems were not previously listed in the NASA Extragalactic Database or Simbad. Besides the interacting galaxy systems, they also found gravitational

lenses, edge-on protoplanetary disks, and overlapping galaxies.

Flamary (2017) trained a simple CNN based on 3 convolutional layers followed by ReLU activation to process ($32 \text{ pixels} \times 32 \text{ images pixels}$) for astronomical image denoising. They compared the results to denoising by the classical Laplacian-regularized Wiener filter and the iterative Richardson-Lucy algorithm, commonly employed in astronomy. Despite the simple architecture, the network has outperformed both traditional methods.

Another interesting use of DL in astronomy is the use of Generative Adversarial Network (GAN) which consists of two CNNs training each other. While the first CNN, also known as the *encoder*, tries to recover the ground truth based on artificially worsened images similar to U-Net, the second CNN, also known as the *decoder*, learns to distinguish between original images and artificially reconstructed images. The aim is to reach a probability close to 0.5 in the decoder part for both of the image types, which means that the decoder can no longer distinguish between original images and artificially recovered images. Like any machine learning approach, however this method has also limitations. Its ability to recover features depends entirely on the training data. To this end, Schawinski et al. (2017) et al. trained a GAN on 4,550 galaxy images at $0.01 < z < 0.02$ from the Sloan Digital Sky Survey where they recovered details in artificially degraded images, which suffer from increased noise and poorer seeing conditions, achieving significantly better results than standard deconvolution techniques. Since it was trained on nearby galaxies, it performs well on similar galaxies captured under comparable conditions. However, applying it to high-redshift galaxies (e.g., $z \sim 2$ or $z \sim 6$) is challenging because their structures differ fundamentally from those at $z \sim 0$. The method also cannot recover extremely subtle features like weak lensing shear, as these distortions are genuinely irretrievable. Additionally, it may struggle with rare objects absent from the training set (Schawinski et al., 2017).

U-Net which will be used in this thesis is a CNN network used to enhance image quality or detect objects. It was first introduced by Ronneberger et al. (2015) as part of research to test the viability of CNNs in the medical detection of cell morphology. They used small dataset of annotated samples and expanded this dataset thorough data augmentation techniques such as elastic transformations, rotations and inversions. The U-Net architecture was able to detect cell geometries (Liu et al., 2025a).

An example application was demonstrated by Zhong et al. (2025) who trained a U-Net to separately extract the flux information of the foreground galaxy (deflector) and background system (source) in galaxy-galaxy strong-lensing events. By combining mock lens components with real sky survey noise, they could accurately model the input image noise, enhance lens modeling, and reconstruct lensing systems from ground-based images within very short time. Furthermore, the U-Net achieved comparable accuracy to traditional lens-modeling tools on space-based lenses and provides reliable photometric redshifts, showing great potential for lens finding, image denoising, and precise modeling in upcoming ground- and space-based surveys (Zhong et al., 2025).

In their recent paper, Liu et al. (2025a) built on the U-Net architecture provided by Ronneberger et al. (2015) with some modifications. They introduced a dropout mechanism

between convolutions to reduce the likelihood of overfitting as the network was trained on a small set of images and partial convolutions were used instead of full convolutions which could eliminate the effects caused by the dropped pixels. Another modification is that the training was completely unsupervised which means that the network had to learn denoising only by inspecting corrupted images. While this approach can be helpful in denoising images when the amount of annotated data is particularly low, it can, however, create abnormal results due to the lack of supervision. While limited more in scope, the authors report that it can significantly reduce noise in solar magnetograms from about 8 Gauss to 2 Gauss (Liu et al., 2025a).

Finally, (Vojtekova et al., 2020) applied a modified U-Net CNN architecture based on (Ronneberger et al., 2015) and (Chen et al., 2018) on HST Wide Field Camera 3 (WFC3) images taken by F555W and F606W captured by the UVIS (UV/visible). The authors tried different loss functions as well as activation functions to evaluate various configurations. Images were consecutively corrupted by scaling them from half to fifth of the original exposure time and various types of noise were added including Photon Noise (PN), Dark Current Noise (DN) and Readout Noise (RN) based on the exposure time and detector specification to mimic realistic images. The U-Net network recovers 95.9% of stars with an average flux error of 2.26%, and the reconstructed images show a signal-to-noise ratio that is 1.63 times higher than the original noisy images, similar to stacking three images (Vojtekova et al., 2020).

Given the potential the U-Net architecture to enhance the quality of images by increasing the SNR, an interesting question is to apply it to galaxies which will be the task of this thesis. A particularly interesting application is the 3D-DASH survey which is the widest survey in the F160W filter corresponding to near-infrared (NIR) range, particularly useful for observing distant, faint objects that might be obscured by dust in the visible light. Using 3D-Drift And Shift method, 3D-DASH expands extragalactic survey area in NIR at HST resolution by an order of magnitude enabling the identification and study of distant galaxies ($z > 2$) that are rare or in short-lived phases of galaxy evolution (Mowla et al., 2022). While 3D-DASH has a wide coverage, images are often shallow images, However, these shallow images can be used to evaluate the performance of Astro U-Net, as the CANDELS survey in the same COSMOS field offers for some of the shallow images deep shots.

Since the results of Vojtekova et al. (2020) look promising, this architecture will be used with some modifications. However, Vojtekova et al. (2020) applied the U-Net on point-like sources such as stars in the visible spectrum. On the one hand, these are often very bright in comparison to galaxies; on the other hand, galaxies are significantly different than stars morphologically. They are extended over an area, are further away and hence fainter which is why they require long exposure times to be captured. These differences necessitate adjustments to the model to ensure its suitability for galaxy observations.

Building on this, the next chapter will introduce the theoretical foundation of convolutional networks, followed by a detailed explanation of the methodology, including the types

of images considered and the process used to generate the training data. The results will be presented in four sections: first, the visual performance of the reconstructed images will be examined, followed by an evaluation based on quantitative metrics. The next two sections will analyze the fluxes of detected objects as well as structural parameters focusing on the Kron radius. In Chapter 5, these findings will be discussed in depth. Finally, the thesis will conclude with a chapter on future work, outlining possible improvements and broader applications of the methodology.

Chapter 2

Convolutional Networks

Just as the processing unit of our brains is the single neuron, the fundamental unit of an Convolutional Neural Network (CNN) is an artificial neuron. Similar to a natural neuron, which connects to multiple neurons and transfers an input signal based on its intensity, an artificial neuron also transforms and passes input signals to subsequent neurons. This process can be mathematically described as a linear transformation:

$$z = \sum_{i=1}^N \omega_i x_i + b \quad (2.1)$$

where i is the index of the input signals to the neuron, ω_i represents the weight assigned to each input, x_i is the input value, and b is the bias term. However, since this is a linear operation, it cannot capture complex patterns or relationships in the data. As linear operations can be mapped back, they offer no information gain. Therefore, non-linearity is introduced via an *activation function* f :

$$h = f(z) \quad (2.2)$$

This function determines whether the accumulated input should be passed to the next layer. While the use of a bias term b is optional, it often enhances model performance. In practice, CNNs use multiple neurons per layer. As a result, the weights must be represented as matrices. For the j^{th} neuron in the k^{th} layer, the process generalizes to:

$$h_j^k = f(z_j^k) = f\left(\sum_{i=1}^N \omega_{i,j} x_i^{k-1} + b_j^k\right) = x_j^k \quad (2.3)$$

where:

- $\omega_{i,j}$ is the weight connecting the i^{th} neuron in layer $k-1$ to the j^{th} neuron in layer k ,
- b_j^k is the bias for the j^{th} neuron in layer k , and
- x_j^k is the neuron's output, serving as input for the next layer.

2.1 Activation Function

The activation function is an essential component of a CNN and it determines whether the accumulated signal from previous layers should be transferred to the subsequent layer by introducing a non-linearity via compression or enhancement of the signal. The non-linearity enables the CNN to learn complex patterns while influencing gradient flow and computational efficiency. Relevant properties of an activation function include differentiability for backpropagation, as well as the range of its output and derivative to prevent vanishing or exploding gradients. The choice of the activation function is a crucial part of an CNN and depends on the research question and the properties of the data. While there are different activation functions and also many custom derivations of the commonly used ones, for the purpose of this work, it suffices to mention the widely used ones:

- **Sigmoid or Logistic Function:**

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (2.4)$$

This function is widely used in various scientific disciplines. Some examples include natural population growth in ecology, diffusion of technology and *know-how* in economy to growth in COVID-19 infections during the pandemic and the Fermi-Dirac distribution of fermions in physics. It is a probability distribution and takes therefore values in the range of $[0, 1]$. Hence, its outputs can be interpreted as the probability of the sample to be a part of a certain class used in the CNN.

- **Hyperbolic Tangent (Tanh) Function:**

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1} \quad (2.5)$$

While both *sigmoid* and *hyperbolic tangent* have similar shapes, the hyperbolic tangent is not a probability distribution and takes values in the range of $[-1, 1]$ with a mean of 0. Therefore, it is best suited for binary classification.

- **Rectified Linear Unit (ReLU):**

$$\text{ReLU}(x) = x^+ = \max(0, x) = \begin{cases} x & \text{if } x > 0, \\ 0 & \text{if } x \leq 0 \end{cases}$$

The ReLU is the most commonly used activation function. One of the most important advantages ReLU offers relative to the sigmoid or hyperbolic tangent is that its derivative is constant. This helps to mitigate the so-called *vanishing gradient problem*. In a CNN, the calculated loss between the actual value, the *ground truth*, and the predicted value is distributed among the layers and then within each layer among the

individual neurons via error propagation. Therefore, a small initial error will vanish altogether till it reaches the early layers. This is evident in sigmoid as the maximum value its derivative can reach is 0.25 as shown in Figure 2.1. Hence, the individual weight corrections in a CNN with many layers and tens if not hundreds of neurons will decrease exponentially with depth. A constant derivative such as that of ReLU can therefore be helpful in such cases. While ReLU offers a solution to the vanishing gradient problem, one of its shortcoming is the projection of negative values to 0 which can be seen in lower left of plot in Figure 2.1. This shuts down all neurons which have returned a single negative value from all subsequent training, effectively reducing the number of trainable neurons.

- **Leaky ReLU:**

$$\text{LeakyReLU}(x) = \begin{cases} x, & x \geq 0 \\ \alpha x, & x < 0 \end{cases}$$

As the name suggests, Leaky ReLU is a modification of the original ReLU. It is used to prevent neurons from shutting down, effectively dying. Leaky ReLU *leaks*, that is to say, it returns a small negative value controlled by α (i.e $\alpha = 0.01$) so that the neuron can be retained in the network for subsequent training.

Due to the advantages Leaky ReLU offers such as the prevention of dead neurons and the mitigation of vanishing gradients through its constant derivative, Leaky ReLU was chosen for the training of our network.

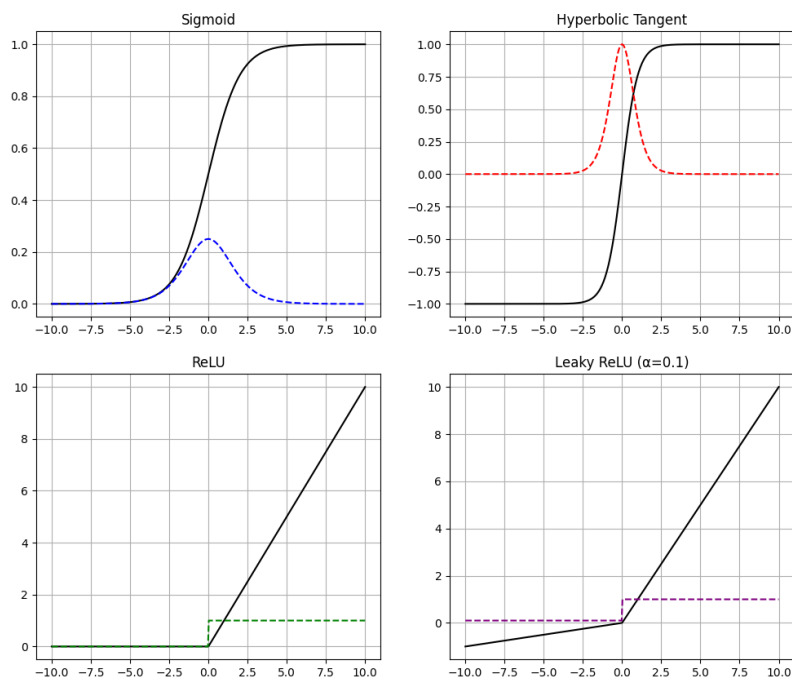


Figure 2.1: Some activation functions (solid lines) with their derivatives shown (dashed lines). The derivative of the sigmoid results in vanishing gradients. ReLU causes dead neurons by projecting every negative value to 0.

2.2 The Loss Function

Another important component of a CNN is the loss function which determines the relation between the prediction and the ground truth. The CNN tries to minimize the loss function through various means. The most common of such methods is the *gradient descent*. The result of the loss function after a successful training epoch will be distributed back to the weights of individual neurons via *back-propagation*. As with most cases, there are different loss functions and the choice thereof depends on the nature of the problem. First, one needs to decide whether the issue is a regression or a classification problem. Regression problems focus on identifying underlying patterns in a dataset, such as data dispersion or common trends, to predict continuous values. In contrast, classification problems aim to distinguish data points by assigning them to predefined categories based on shared characteristics. Some of the commonly used loss functions for regression problems are **mean square error (MSE)** (also known as L2 distance) and **mean absolute error (MAE)** (also known as L1 distance), as these pertain to the relative or absolute distance between the prediction and the ground truth. Classification problems, on the other hand, require some sort of

probability which is why **probability distributions** or **cross-entropy functions** are used here. Table 2.2 summarizes the most widely used loss functions and their usage.

Table 2.1: Common Loss Functions and Their Uses

Loss Function	Formula	Use Case
Mean Absolute Error (MAE)	$MAE = \frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $	Regression
Mean Squared Error (MSE)	$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$	Regression
Binary Cross-Entropy	$L = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$	Binary Classification
Categorical Cross-Entropy	$L = -\sum_{i=1}^n \sum_{j=1}^c y_{ij} \log(\hat{y}_{ij})$	Multi-Class Classification

As the aim of this project is to improve the quality of shallow astronomical images and not to classify objects, MAE and MSE are the only possible candidates out of the listed candidates above. While it is possible to use either of these, Vojtekova et al. (2020) have showed in their work that MAE loss was not appropriate for the Astro U-Net problem. Furthermore, (Zhao et al., 2017) demonstrate that MSE loss is more prone to getting stuck in local minima, while MAE loss tends to reach the minima more effectively. It is also worth noting that MSE loss is more sensitive to large errors. When the data distribution is likely to contain outliers, MAE loss is generally a better choice due to its greater robustness. Due to the above mentioned considerations, MAE will be used as the loss function.

Optimization

The weights should be updated using the distributed loss from the loss function. This is where optimization functions are used. These decide how the weights should be updated. The most widely known and the basic one is the *gradient descent*. The idea is very simple and intuitive in the sense that the update of the weights follow the steepest curve. To accomplish this, the loss function will be partially differentiated with respect to the respective ω_i which maximizes the said function with the said weight. This value is then multiplied by the learning rate α and subtracted from ω_i . The subtraction ensures that the weights are updated accordingly as a positive partial derivative shows the magnitude of the slope *away* from the minimum but not *towards* it. This can be summarized in:

$$\omega^{(t+1)} = \omega^{(t)} + \Delta\omega^{(t)} \quad (2.6)$$

$$\Delta\omega^{(t)} = -\alpha \sum_{i=1}^N \frac{\partial L(x)}{\partial \omega^i} \quad (2.7)$$

Where α stands for the learning rate, t for the epoch, L for the loss function and x for the dataset. One of the biggest disadvantages of a plain gradient descent is that it takes the derivative at the end of each epoch, since it accumulates the loss from each training example which can be slow depending on the number of training examples. However, since L is the sum of individual losses and the differentiation operation is a linear one, the

gradient descent has to calculate N single gradients at the end of each epoch, which makes the updating process computationally very slow and expensive. Furthermore, the whole dataset during this process has to be stored in the memory which is often times not possible. One advantage the plain gradient descent offers, however, is the guarantee of convergence as long as the loss function is a convex one. However, as no jumps are possible, the gradient descent might end up in a local minimum instead of the global minimum. Hence, plain gradient descent is not a good approach.

Stochastic gradient descent (SGD) is another method that aims to solve these issues. It is given by the following formula:

$$\Delta\omega^{(t)} = -\alpha \frac{\partial L(x)}{\partial \omega^i} \quad (2.8)$$

The difference between (2.7) and (2.8) is that the former uses the accumulated loss of the whole dataset to update the weights, whereas the latter uses a single random training sample x_i and $\left(\frac{\partial L}{\partial \omega_i}\right)$ gradient and updates the weights after each training sample x_i . It therefore skips the redundancy of batch gradient descent by updating the model one step at a time. Because it updates so frequently, the objective function fluctuates a lot, unlike batch gradient descent, which steadily moves toward a minimum. This randomness can be both a strength and a weakness. On the one hand, it helps SGD escape bad local minima and potentially find better ones, but on the other hand, it makes precise convergence harder, since it keeps overshooting. However, if the learning rate is gradually reduced, SGD behaves more like batch gradient descent and can reliably reach a local or global minimum, depending on the type of optimization problem (Ruder, 2016). This overshooting can be reduced by introducing a *momentum* term γ also known as the first moment:

$$\Delta\omega^{(t)} = \gamma\Delta\omega^{(t-1)} - \alpha \frac{\partial L}{\partial \omega_i} \quad (2.9)$$

where $\Delta\omega^{(t-1)}$ is the change in weight of the previous epoch and γ is the momentum coefficient, which is usually set up to 0.9. The momentum term becomes more important if the loss function is multidimensional, since the function might have a different behaviour in other dimensions around the searched minimum. By *remembering* the previous value, the momentum term provides the SGD a direction to descend to. As Figure 2.2 shows, the added momentum helps to reduce the oscillations around the minimum in comparison to left figure where the oscillations are bigger.

There are further optimizers such as **Adagrad** (Duchi et al., 2011), **Adadelta** (Zeiler, 2012) and **RMSprop** (Tieleman and Hinton, 2012). The last one uses squared gradients also known as the second moments to adjust the learning rate for each parameter, effectively reducing the step size when the gradients are large and increasing it when the gradients are small (Tieleman and Hinton, 2012).

To sum up, the momentum speeds up training by providing a direction to SGD while squared gradients adapt the learning rate but may struggle in some cases due to its lack of

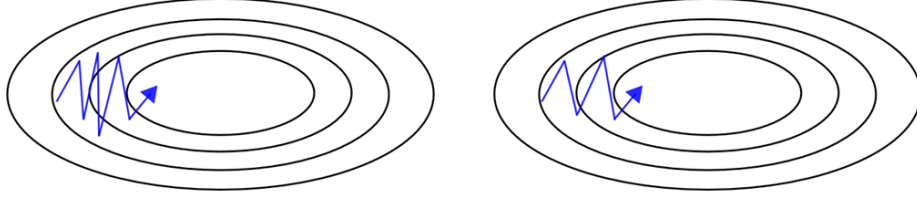


Figure 2.2: Left: SGD without momentum. Right: SGD with momentum (Ruder, 2016) Oscillations are bigger and more common in the SGD without the momentum than with momentum.

momentum. **Adaptive Moment Estimation** also known as **Adam** expands RMSprop by adding a momentum term, also known as the first moment which leads to combination of both adaptive learning rate and faster convergence due to momentum. It is memory efficient and works well with noisy data and deep CNNs (Kingma and Ba, 2014). Due to the said advantages relative to the simple SGD with momentum and RMSprop and its well behaviour towards noisy data which astronomical images usually are, Adam will be used as the optimizer for Astro U-Net. The weight update using Adam is calculated in the following way:

$$m^{(t)} = \beta_1 m^{(t-1)} + (1 - \beta_1) g^{(t)} \quad (2.10)$$

$$v^{(t)} = \beta_2 v^{(t-1)} + (1 - \beta_2) (g^{(t)})^2 \quad (2.11)$$

$$g^{(t)} = \frac{\partial L}{\partial \omega_i} \quad (2.12)$$

$$\omega^{(t)} = \omega^{(t-1)} - \frac{\alpha m^{(t)}}{\sqrt{v^{(t)}} + \epsilon} \quad (2.13)$$

where $m^{(t)}$ and $v^{(t)}$ are the first and second moment estimates and β_1 and β_2 and their corresponding decay rates respectively.

2.3 Convolutional Layer

The convolutional layer is the main pillar of a CNN. It is similar to RNN in that its input, flattened to be a feature vector, must go through an activation function. However, this flattening of the input data causes the loss of spatial or temporal information in the data for instance when the data is a time series or an image. Pixels in an image that are close together, are more likely to have similar values than those that are further apart. Hence, it is often redundant to run the whole image through an ordinary neural network considering that the dimension of the feature vector scales with $(H \times W)$. This is where the convolution

becomes beneficial. It shrinks the size of the input data by *sliding* a window on the spatially encoded data.

Mathematically, the convolution operation ($\star$) can be expressed as:

$$G = F \star H \quad (2.14)$$

$$G(x') = \int_{-\infty}^{\infty} F(x)H(x' - x)dx \quad (2.15)$$

where F is the signal, H the filter and G the convolved signal. For discrete data in 2 dimensions such as images, the convolution operation can be written as:

$$g'_{ij} = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f_{i+m,j+n} \cdot h_{m,n} \quad (2.16)$$

where i and j denote the location in the convolved two dimensional data, $f(i, j)$ the signal or the input data and $h(m, n)$ the kernel or the filter and lastly M and N the dimensions of sliding window or the filter. Figure 2.3 demonstrates how this works in practice.

As seen in Figure 2.3 convolving the image with an image reduces the output size. The figure also shows an important aspect of convolution which is *padding*. Padding is the process of adding a frame around the data. There are different methods such as adding a **constant padding** which is usually 0, **replicate padding** by using the boundary values or **reflect padding** by mirroring the data values in the outward direction. Padding can be useful when boundaries are important or when the input or output dimensions must match. Padding will be used in Astro U-Net as the aim is to recover the ground truth image from a corrupted one. Therefore, the input and output shapes must match. A constant padding with a value of 0 will be used as adding a positive constant would effectively create extra noise and mirroring would create unreal sources if there should be some near the edges.

Another important aspect of a convolution is the *stride* which determines how many pixels the filter should move by at each step. Choosing a low stride will increase the computational load without necessarily adding significant value, as overlapping pixels in the output may contain redundant information. On the other hand, using a high stride may miss out on spatial details by skipping parts of the image. Putting all these together, the output dimensions of a single CNN layer is given by the following formula:

$$H_O = \frac{H_I + 2P - F}{S} + 1 \quad (2.17)$$

$$W_O = \frac{W_I + 2P - F}{S} + 1 \quad (2.18)$$

$$D_O = N_F \quad (2.19)$$

where:

- P stands for padding,

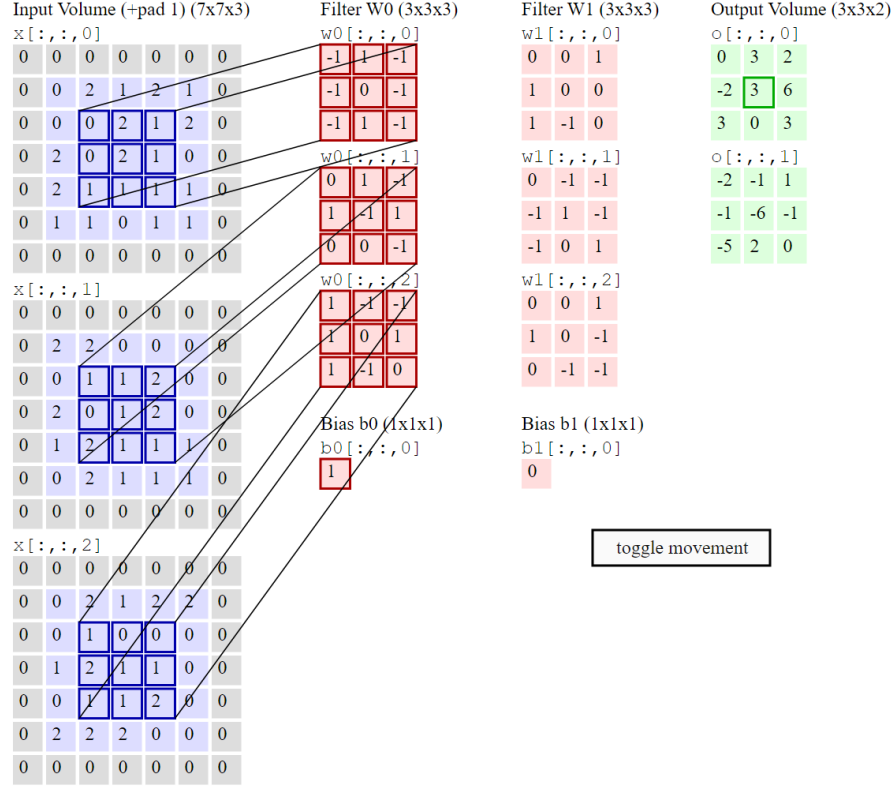


Figure 2.3: The lighter blue frame is the image data surrounded by a single line of null padding. The rows correspond to the last dimension D of $(H \times W \times D)$, also known as the color channels. Each data channel is slid over by the channel of a filter on the same level. Hence $D_{\text{data}} = D_{\text{filter}}$. The resulting matrices will be laid over and summed so that a single matrix per filter is attained. Using two different filters W_0 and W_1 leads to two different outputs: $O[:, :, 0]$ and $O[:, :, 1]$. An interactive map can be found at: (CS231n Stanford, last accessed on March 4, 2025)

- S denotes the stride,
- and D_O and N_F the depth or the number of channels of the output and the number of filter respectively.

While it is theoretically possible for stride and padding to change across height and width (i.e. *rectangles*), they are almost always the same (i.e. *squares*), at least in the state-of-the-art libraries.

2.4 Pooling Layer

In a usual CNN, the convolutional layer is often followed by a pooling layer. The goal of pooling layers is similar to convolutional layers, that is, the reduction of dimensionality while gaining important spatial information. Mathematically, the difference between a convolutional and pooling operation is that the former *convolves* the function f with another function h such that:

$$g(x, y) = f(x - x', y - y') \star h(x', y') \quad (2.20)$$

while the latter *applies* the function h to the function f in some neighbourhood $\mathcal{N}$:

$$g(x, y) = h(\{f(x', y') : x' \in \mathcal{N}(x), y' \in \mathcal{N}(y)\}) \quad (2.21)$$

Typically, the applied function is either the maximum value $\max(\mathcal{N}(x, y))$ or the average. Visually, this represents that the sliding filter does not take the matrix multiplication of the area but calculate either the maximum or the average value. Maxpooling will be used as part of Astro U-Net as this is the pixel which is the least contaminated with noise.

Since the operations in a CNN and a pooling layer are *mechanically* the same, the same dimension calculations of the convolutional layer (2.19) is also valid for the pooling layer. Another advantage of using a pooling layer even in the case of *same* padding when the input and the output sizes are the same, is that it introduces translation invariance by shifting the location of the maximum or the feature (Dumoulin and Visin, 2016).

2.5 Deconvolutional Layer

Deconvolutional layers, also known as transpose convolutional layers, are not typically part of a CNN, unless the task involves semantic segmentation, image classification, super-resolution, that is, the improvement of resolution or image reconstruction (Dumoulin and Visin, 2016). It can be seen as *undoing* what the convolutional or pooling layer has done by expanding the dimensions. In core, this involves an interpolation. Therefore, it can be seen as convolution or pooling going in the reversed direction.

One of the problem of upsampling is that the spatial information gets lost during a pooling and downsampling operation. A possible remedy is the storing the location of the maximum in a sparse matrix during the downsampling operation and using this information during the upsampling operation to project the maximum values back to their original locations as successfully demonstrated by (Zeiler et al., 2011), (Zeiler and Fergus, 2014) and (Noh et al., 2015). This is shown in Figure 2.4. Switch variables shown in the pooling example act as the sparse matrix and store the location of the maximum which are then used in the subsequent unpooling operation to reproject the value back to its original position. It is also possible to stack multiple deconvolutional layers one after another with an activation function so that the weights of the interpolations can be trained (Shelhamer et al., 2017). This later strategy will be used in Astro U-Net.

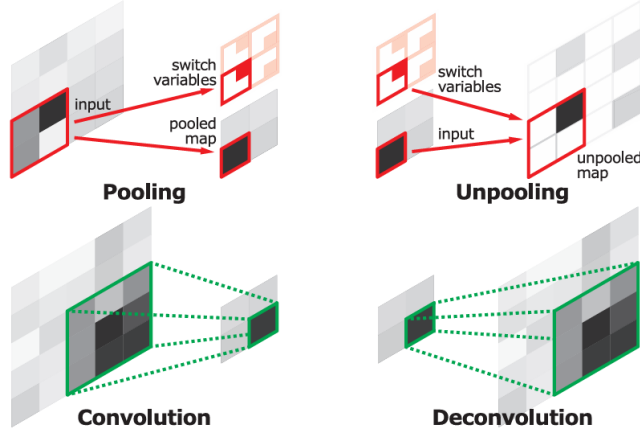


Figure 2.4: Comparison of Down- and Upsampling. Switch parameters, i.e. the locations are stored in the pooling to be later used in the unpooling (Noh et al., 2015)

2.6 Miscellaneous Layers

- **Dropout** : is the process of randomly turning off some neurons during training to prevent the network from relying too much on specific ones, as it decreases the possibility of the network to memorize the data. This makes the model more robust, though it can slow down training.
- **Batch Normalization**: The input data to neural networks are often stacked together to increase the efficiency by calculating the activation and loss functions across such chunks of data, also known as **batches**. Batch normalization standardizes outputs of each layer by normalizing them across these. This reduces variations in activation distributions, stabilizes training, and helps the network converge faster.

2.7 U-Net

The name *U-Net* stems from the fact that it has a U shape architecture, meaning the upsampling or contracting path mirrors the downsampling or expansive path. Figure 2.5 demonstrates an example U-Net architecture. The left half corresponds to the contracting path. The consecutive application of convolutions followed by max pooling decreases the size of the image by half, as 2x2 pooling is used, while increasing the number of channels or depth by the same ratio at each level. Each block contains two convolutions followed by ReLU. The right half reverses the process performed by the left path. At each step, the transpose convolution (or upsampling) increases the spatial resolution, projecting the accumulated features onto a canvas four times the size of the layer. This new layer is

concatenated with the corresponding layer on the left side of the network to regain spatial information. This is followed by two additional convolution and ReLU blocks, which allow the network to learn how to interpolate the features correctly.

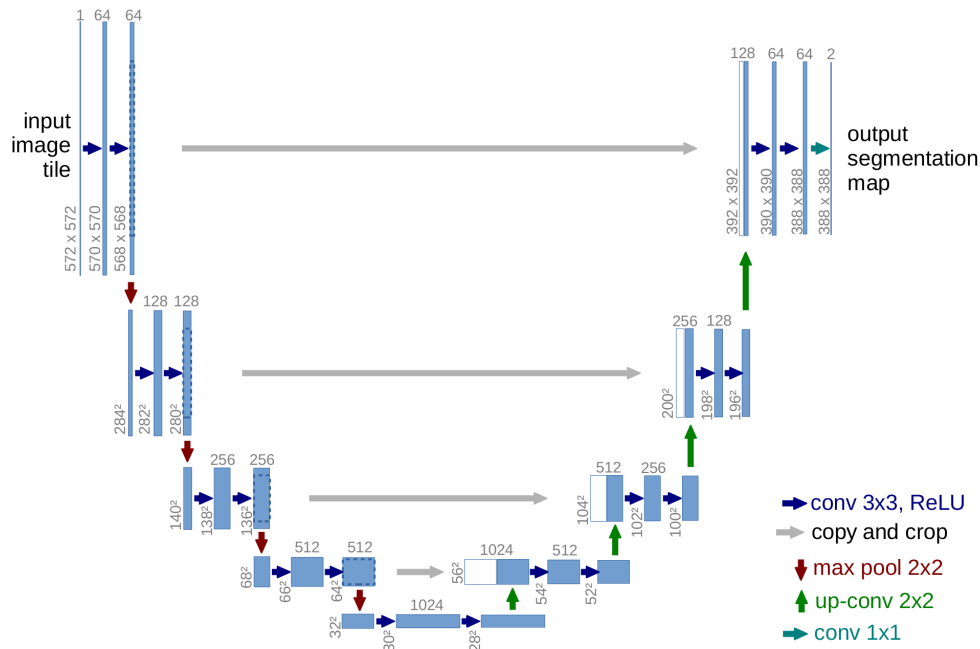


Figure 2.5: An exemplary U-Net architecture used in (Ronneberger et al., 2015). contracting (left) half of the architecture mirrors the expansive path (right). Concatenating the results from the downsampling part with the upsampling part enables the network to regain the spatial information lost during the pooling.

An example application was shown by Chen et al. (2018) who used a U-Net architecture in low light imaging with short-exposure times which is challenging due to low photon count and poor SNR. Short-exposure images are noisy, while long exposures risk blur and impracticality. Existing denoising and enhancement methods struggle in extreme conditions, like video-rate night imaging (Chen et al., 2018). The authors trained a fully convolutional network based on a U-Net architecture on raw sensor data to compare its performance relative to traditional pipelines which perform poorly under such settings. Figure 2.6 demonstrates how U-Net architecture outperforms traditional image enhancements. As the figure demonstrates under very short exposure settings $\sim \frac{1}{10}$ seconds, U-Net (right) can enhance the image significantly better than traditional enhancement methods (middle).

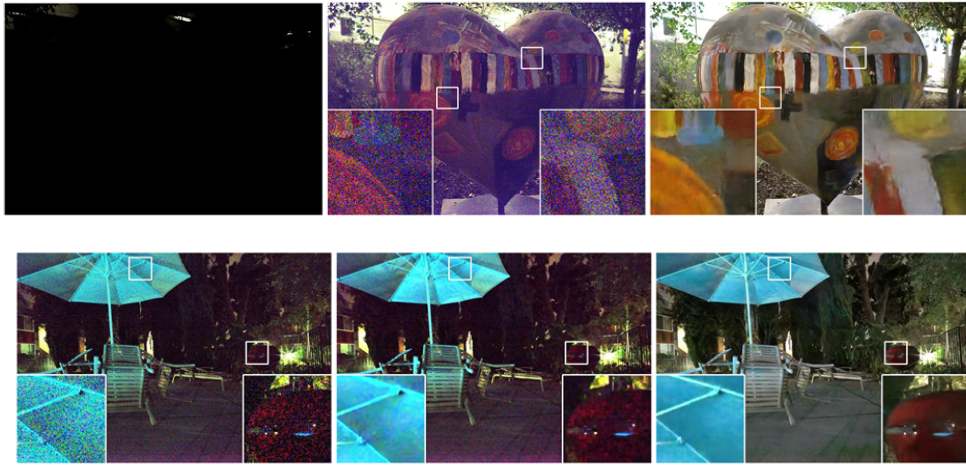


Figure 2.6: The performance of U-Net (right) in comparison to the traditional image enhancement methods (middle) and raw lens data (left) under very small exposure time $\sim \frac{1}{10}$ seconds and in low lighting (Chen et al., 2018)

Chapter 3

Methodology

3.1 U-Net Architecture

The U-Net architecture used by Vojtekova et al. (2020) follows the same outline as the original architecture of (Ronneberger et al., 2015). There are five layers in total. Each layer consisting of two convolutions and single maxpooling. However, (Vojtekova et al., 2020) replaces ReLU with LeakyReLU to account for dead neurons. Like (Ronneberger et al., 2015) the kernel size for convolutions is given by (3 x 3) while those of pooling and transpose convolutions are (2 x 2). The code provided by (Vojtekova et al., 2020) can be found at GitHub ¹. In our work, we built on the architecture provided in Vojtekova et al., 2020 and made the following changes which can be investigated in Figure 3.1:

1. Each convolution before the activation function *LeakyReLU* is followed by *batch normalization*. CNNs tend to perform better and converge faster when the input data is normalized beforehand. Furthermore, (Zhang et al., 2017) has shown that adding batch normalization boosts the performance of denoising and shortens training time. Hence, we choose to integrate this into the modified U-Net.
2. A *dropout* layer with a default ratio of 0.25 has been added before each max-pooling operation. This means that 25% of the neurons are randomly deactivated at each layer during training. Although dropout increases training time, it effectively mitigates overfitting by preventing the model from fitting to noise. (Liu et al., 2025a), specifically, demonstrated that incorporating dropout in a modified Astro U-Net significantly reduced overfitting.

¹Vojtekova 2020, AstroUnets, last accessed on 14.03.2025

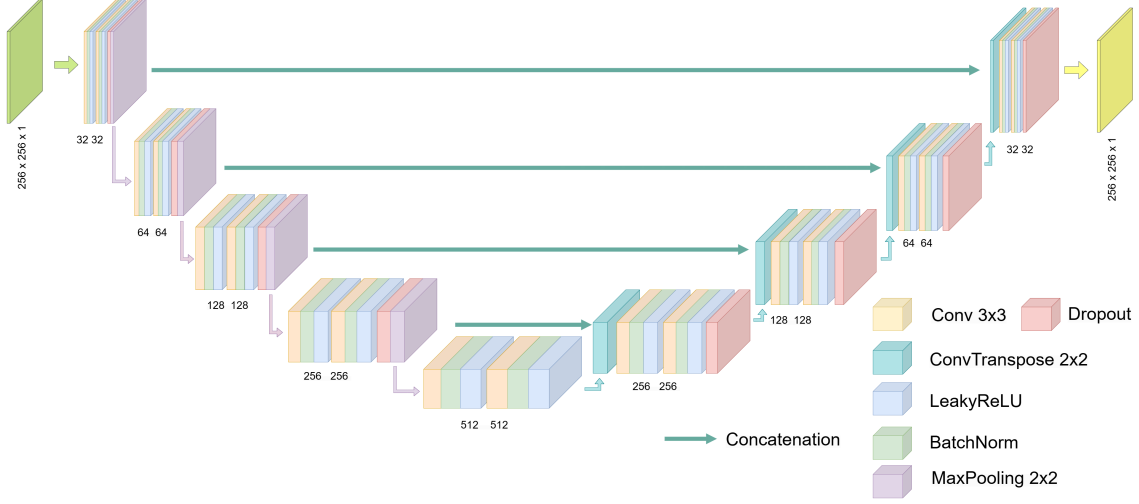


Figure 3.1: Modified Astro U-Net Architecture. It follows a symmetric structure with skip connections to help preserve spatial details. The encoder (left side) gradually reduces the spatial resolution of the input ($256 \times 256 \rightarrow 128 \times 128 \rightarrow 64 \times 64$, etc.) while increasing the number of feature channels ($32 \rightarrow 64 \rightarrow 128$), shown under each layer. At the bottleneck, the network captures the most compressed representation of the data before passing it to the decoder (right side), which uses transpose convolutions to upgrade and restore the original image resolution. The use of dropout prevents overfitting and batch normalization allows a smoother convergence.

3.2 Data

The images used in the training of the network stem from observations taken by the Hubble Space Telescope (HST) between 2010 and 2024 using the Wide Field Camera 3 (WFC3) which is the most advanced camera mounted on the HST to capture light across a wide spectrum, from ultraviolet to near-infrared wavelengths (200 to 1700 nm). It combines two optical/ultraviolet sensors with a near-infrared detector, making it versatile for various astrophysical studies. The instrument supports different types of imaging through a variety of filters. This wide spectrum is divided between two main channels:²:

1. **Infrared (IR) Channel:** Uses a specialized detector for near-infrared imaging, covering a field of 123×136 arcseconds with a pixel size of 0.13 arcseconds. It operates at 145 K and handles faint light well, with minimal noise (Marinelli and Green, 2024).

²WFC3 Instrument Design, last accessed on 14.03.2025

2. **Ultraviolet/Visible (UVIS) Channel:** Optimized for Ultraviolet and visible imaging with two CCD sensors. It captures a broader field of 162 x 162 arcseconds with a finer pixel size of 0.04 arcseconds. It operates at 190 K, offering excellent sensitivity and low noise (Marinelli and Green, 2024).

From these two channels, this thesis will concentrate on the images captured by the IR channel as the goal is to apply Astro U-Net to deeper galaxies. Due to the expansion of the universe, light from other galaxies are stretched along with the space which causes them to shift to longer infrared wavelengths also known as the redshift. Therefore, the IR channel is widely used in the study of early universe, super massive black holes (Cammelli et al., 2025), the relation between morphology and star formations (Ge et al., 2024), spectra of far away galaxies and objects (Brammer et al., 2012) and their evolution (Mowla et al., 2022).

There are numerous filters on WFC3 for the IR channel, ranging from narrow to wide spectra. The F160W WideBand Filter was chosen as it provides wide coverage between 14,000 Å and 17,000 Å³. This is advantageous, on the one hand due to its broad spectral range, and on the other hand, the 3D-DASH and CANDELS (Grogin et al., 2011) (Koekemoer et al., 2011) surveys provide shallow and deep imagery using these filters, respectively.

3.2.1 Data Selection and Download

To train a model which replicates reliable results, it must be trained on a dataset that offers various different structures on varying exposure times. Furthermore, the number of images in the training set must be high enough to reduce the possibility of overfitting. Therefore, some statistics were run on the metadata, which was then filtered, so that only qualifying images were downloaded. To this end, the MAST portal⁴ was queried using `MastMissions` function of the `astroquery` package. The MAST portal can be queried based on different filters. For the purpose of this thesis, corresponding filters were given⁵ which yielded 12,868 different observations. These were filtered based on aperture parameter (`sci_aper_1234`) which determines the placement of the target on the detector. Following apertures were selected:

- **IR:** Places targets at the detector’s optimal center; its default position may be adjusted based on detector performance. Best for small targets.
- **IR-FIX:** Fixed at the detector’s geometric center, unaffected by detector changes. Suitable for positioning extended scenes.

³HST WFC3 Instrument Handbook: IR F160W, last accessed on 14.03.2025

⁴MAST Portal, last accessed on 14.03.2025

⁵{`sci_spec_1234` : 'F160W,F160W;*,*;F160W;*,F160W;*', `sci_obs_type` : 'image', `sci_aec` : 'S', `sci_instrume` : 'wfc3'}

- **IR-UVIS, IR-UVIS-FIX, IR-UVIS-CENTER:** Align with corresponding UVIS apertures to avoid small angle maneuvers between IR and UVIS exposures.

Furthermore, five apertures are designed for IR grisms (G102 and G141). They are named **GRISMmm**, where mm denotes the field of view (FOV) size: 1024 (full frame), 512, 256, 128, and 64. Only the full frame was selected. More information about different apertures and their use cases can be found at the HST User Documentation ⁶.

Resulting metadata was then filtered to ensure a good coverage with respect to exposure time. 3,000 images have been selected in total. Following steps are used thereby:

1. Only images with exposure times between 200 and 20,000 seconds have been selected. The reason for choosing 200 seconds as the lower limit was to ensure that the artificially corrupted images during noise addition in subsequent stages retained enough structure for meaningful degradation and analysis.
2. If fewer than **3,000 images** remained after filtering, all available images that passed the previous step were included without further restrictions.
3. Otherwise, the selection process was as follows:
 - The metadata was shuffled, and images with unique exposure times were chosen first.
 - If this still resulted in fewer than 3,000 images, the metadata was sorted by exposure time, and additional images were selected based on a normal distribution of indices.
 - This process was repeated until exactly 3,000 images were selected.

Keeping at least one occurrence of each unique exposure time and selecting the remaining images from a normal distribution of indices based on the sorted exposure time data ensures broad coverage across exposure times. This approach maintains a representative training and test set with respect to the available surveys on MAST. Furthermore, two different models with the same architecture were trained on two different datasets: one using the aforementioned general dataset and another specifically focused on PI data. To construct the latter, only observations conducted in the following surveys were considered: *FABER*, *LOTZ*, *COE*, and *POSTMAN*. Aside from filtering the metadata based on these surveys, all other filtering criteria remained the same. Training two separate models —one on a general dataset and one on a specialized PI dataset allows for a comparative analysis of how data-dependent the model’s performance is.

Figure 3.2 shows the exposure time distribution of WFC3 images with F160W on MAST (left), selected images for the general case model (middle) and selected images for the model that was trained on selected PI galaxy data only. The figure shows, that

⁶HST User Documentation, last accessed on 14.03.2025

there are more shallow images than deeper images in MAST which should be of no surprise. Furthermore, the middle histogram demonstrates, that there is almost an even distribution of selected images for the general model in the range of 200 to 1,800 seconds. This must be attributed to the first step of the selection, namely, the selection of at least one image per individual exposure time. Other than that, the distributions on MAST and selected images for the general case follow similar trajectories. Selected surveys on the other hand has a different distribution pattern which must be due to the smaller subset of projects that were considered and hence, due to the more homogeneous nature of the dataset. Also, the number of images selected in the latter ($N = 1628$) was less than the amount of 3,000.

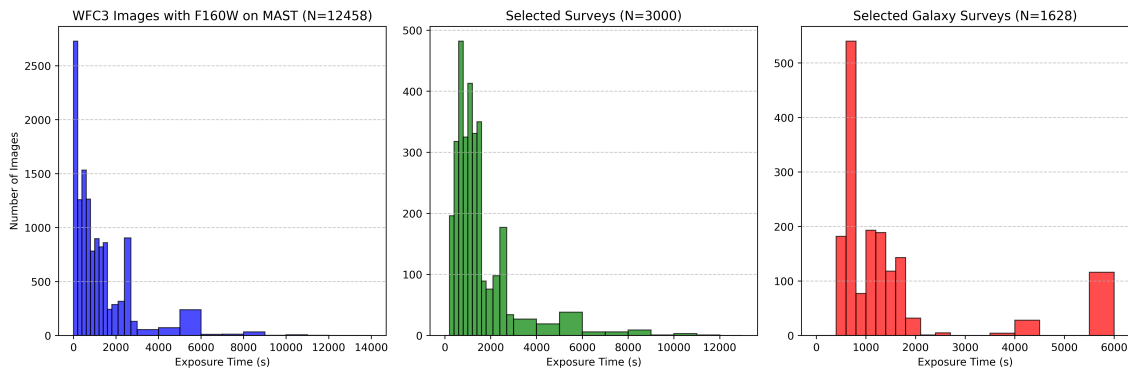


Figure 3.2: Exposure time distributions for WFC3 F160 W images on MAST (left), the general dataset (middle), and the PI dataset (right). The MAST distribution shows a dominance of shallow images. For the general dataset, the middle histogram reveals a nearly uniform distribution of selected exposure times between 200 and 1,800 seconds, a result of ensuring at least one image per unique exposure time. The PI dataset, however, shows a significantly different distribution, driven by the lower number of surveys considered here illustrating the impact of data selection on the final exposure time distribution.

Having filtered images, they were downloaded, a training-test split of 0.7 was chosen and the selected images were grouped either under the training or the test set based on this ratio. A ratio of 0.7 allows a reasonable validation while keeping the majority of the images for the training purpose. Furthermore, some important statistics about the images were calculated and saved in a file during the downloading process so as not to calculate them at every epoch during training. These statistics include the *mean*, *median* and the *standard deviation* of the background as well as of detected light sources. To this end, `sigma_clipped_stats` and `SigmaClip` from `astropy` as well as `detect_threshold`, `detect_sources` and `circular_footprint` functions of `photutils` libraries were used. The parameters used for this purpose are:

- `'sigma': 2`

- 'nsigma': 3
- 'footprint_radius': 5
- 'npixels': 5
- 'maxiters': 10

Aside from `sigma` parameter which was set to 2 instead of the default 3, all other parameters were kept with their default values. The reasoning for choosing a smaller sigma value than the default was to ensure that no boundary pixels around bright sources skew the background calculation. While losing a subset of background pixels during the sigma clipping was not relevant in the calculation of the overall background and its standard deviation, keeping some of the PSF pixels at the boundaries bears the potential to increase the error of background which was avoided by choosing a smaller sigma value.

The mean (μ) and standard deviation (σ) of the background of the downloaded images were plotted to better understand the noise distribution. Figure 3.3 shows the distributions of both values for the general dataset as well as the selected surveys. In both datasets, the majority of the images have background values around $0.5 \frac{\text{electrons}}{\text{second}}$, with the frequency of occurrences decreasing exponentially as the values move away from this point. However, the general dataset shows a wider spread, whereas the selected surveys more tightly clustered. This can be explained by the smaller number of programs with more homogeneous observations in a handful of fields in the latter. A similar behavior is observed for the background error, though the values are about an order of magnitude smaller.

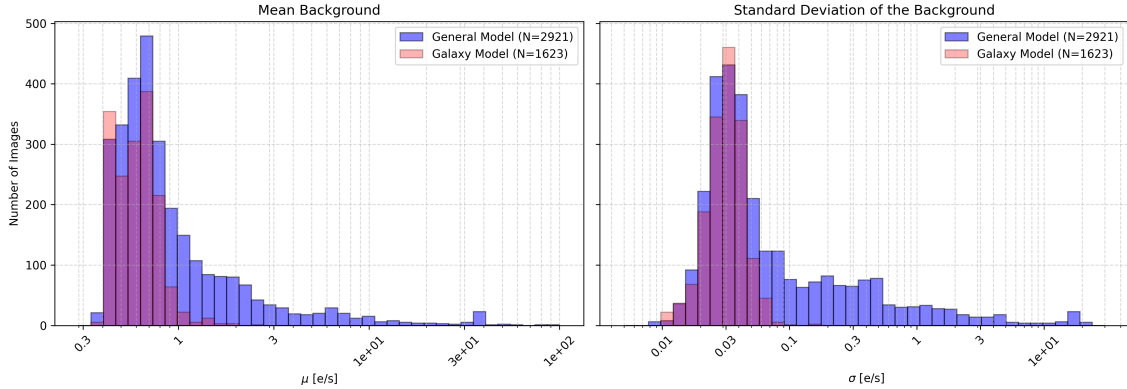


Figure 3.3: The Distribution of mean background (left), the distribution of the standard deviation of the background (right). Note that the $N_{general}$ decreased from 3,000 to 2,921 while N_{galaxy} from 1,628 to 1,623. This must be due to `sigma_clipping`. These images will be excluded from further analysis.

After the images were downloaded and relevant statistics were calculated, the images were cut into (256 pixels x 256 pixels) patches and where the image dimensions required it, padded around the edges to ensure compatibility with the model.

3.2.2 Data Augmentation

The quality of a model's performance on real data is closely tied to the quality of the data used for training. Therefore, care must be taken to ensure that the corrupted images provided to the model during training closely mimic real-life data at short exposure times. However, a relation between exposure time and noise must be found before further steps can be taken.

The noise in the astronomical imaging comes mostly from three different sources which can be formulated in the following way (Hainaut, 2005):

$$\sigma_{total}^2 = \sigma_{PN}^2 + \sigma_{DK}^2 + \sigma_{RN}^2 \quad (3.1)$$

where PN, DK and RN stand for the photon, dark current and readout noise respectively. DN is a result of thermodynamics and caused by the motion of electrons within the sensors. It increases linearly with time as more electrons are accumulated. RN arises when the accumulated signal in terms of photons are registered as electron current and is considered constant for a particular detector. PN, on the other hand is an intrinsic property of photons and is due to the discrete quantized nature of photons as particles. They arrive randomly at the detector which causes this type of noise. Due to its discreteness, it can be explained by the Poisson statistics:

$$N(t) = \frac{\lambda^t \cdot e^{-\lambda}}{t!} \quad (3.2)$$

where λ is the rate of arrivals of photon for some interval t or flux. The Poisson distribution evolves into a Gaussian distribution in the case of high λ . In longer exposure times, the PN regime will dominate in (3.1) such that $\sigma_{total}^2 \approx \sigma_{PN}^2$ which will be assumed in this thesis.

The Poisson distribution is explained by the single parameter λ where $\mu = \sigma^2 = \lambda$. The expected number of photons N at the detector is:

$$E[N] = \lambda T \quad (3.3)$$

As our data is given in $\left(\frac{e^-}{s}\right)$, a time dependent σ_λ is required to augment samples. This can be derived by inverting 3.3 and then propagating the error as follows:

$$\lambda = \frac{N}{T} \quad (3.4)$$

$$\Rightarrow \Delta\lambda = \frac{N}{\sqrt{T}} = \sigma_\lambda \quad (3.5)$$

$$\Rightarrow \sigma^2 \propto T \quad (3.6)$$

Using this one arrives at:

$$\Rightarrow \frac{\sigma_{T_1}^2}{\sigma_{T_2}^2} = \frac{T_2}{T_1} \quad (3.7)$$

Having derived a relation between the flux error and the exposure time, one can create the augmented data samples as follows:

1. The highest variance in the dataset was assumed to come from the image with the shortest exposure time, denoted as $\min_ \sigma^2$.
2. Starting from the variance of each patch, σ_{patch}^2 , the variance of the augmented patch, $\sigma_{patch_{aug}}^2 = a \cdot 10^b$, was calculated by increasing b until it matched the exponent of $\min_ \sigma^2$. For each magnitude, three such values of $\sigma_{patch_{aug}}^2$ were generated, corresponding to three different values of a . Each integer increase in b will result in an exposure time decrease of a magnitude while each a will cover approximately an a decrease of 0.3 magnitude in exposure time.
3. Using the relation (3.7) the expected new exposure time of the augmented patch, t_{aug} , was calculated. Patches with t_{aug} below 90 seconds were filtered out.
4. A patch signal ratio, $\rho = \frac{\mu_{sn}^p}{\mu_{img}^{img}}$, was computed, where μ_{sn}^p represents the mean signal of the respective patch and μ_{sn}^{img} is the mean signal of the corresponding image. These values were read directly from the statistics recorded during the download.
5. A table was created with the computed values. The table was sorted by image ID to group patches from the same image together and then sorted in descending order based on ρ . Using ρ , priority was given to patches with the highest signal, ensuring that they were selected first from each image.
6. A loop iterated through the table, selecting, for each image, the patch with the highest ρ . From the three possible values of $\sigma_{patch_{aug}}^2$ corresponding to that ρ , the loop selected the largest base a while ensuring that the exponent b was as close as possible to that of the ground truth σ_{patch}^2 . The selected values were stored in a new table and removed from the original one.
7. Steps 5 and 6 were repeated until 3,000 patches with different $\sigma_{patch_{aug}}^2$ were collected. However, instead of always selecting the highest ρ , a random selection was introduced to increase diversity in the dataset.

These steps ensure that at least the patch with the highest signal from each image is always selected. If additional slots are available, new patches are included, and the exposure ratio increases in steps of approximately 0.3 per magnitude. Furthermore, during training, a

random number, η , between 0 and 3 determines the rotation of the image by $(\eta \cdot 90^\circ)$ to introduce diversity and rotation invariance.

Figure 3.4 shows the distribution of exposure time ratios between the ground truth and the corrupted images for both the general data and the selected PI surveys. Both distributions follow a downward trend. However, the distribution of the PI dataset reveals an irregularity between the values of 5 and 15. The distribution hits a minimum around 10 and then slowly rises to 15, where the count matches the value at 5. A possible explanation for this is the peak around 700-800 seconds in the left plot of Figure 3.2, which displays the exposure time distributions of the samples. More than one-third of the selected images are represented by this curve. These images cannot be corrupted beyond 10^{th} of their original exposure time before the duration drops below 70 seconds. However, as shown in the figure, there are also images above this threshold that can still be corrupted further, so the *hit* and subsequent *rebound* makes sense.

One might be tempted to think that this hit could skew the results. However, as Figure 3.5 illustrates on the left, the number of patches considered per image follows a broad pyramidal shape. This means that at least two, if not three, samples are included per image. The number of images contributing more than three patches seems, visually at least, to be quite rare. Interestingly, the oscillations in the general dataset appear to be larger than in the selected surveys, even though the general data shows a regular decline in Figure 3.4. This is likely due to the higher number of different surveys involved in the general dataset, leading to a more heterogeneous nature.

Having calculated the noise levels of augmented images, the network was run. The augmented images were created on the fly by adding the calculated noise level on top of the original image ⁷ with a mean around 0. As the Poisson distribution evolves into the normal distribution with larger fluxes, this was assumed for the ease of calculation. In the cases, where this has resulted in negative pixels, the image was shifted by the minimum value.

As .FITS images can span multiple images, the question arose whether it was appropriate to scale the images before providing them to the network. CNNs are known to work when the input data is normalized. Since this was hard to ascertain beforehand, the network was run with and without scaling to compare the performance. Different scaling options are considered such as *min-max*, *log* and *z-scaling*. Z-scaling works best when the input data follows a normal distribution which the noise in astronomical images is assumed to be. After some trials and preliminary results, z-scaling was chosen as this looked visually more adequate. Z-scaling also known as the z-score is given by:

$$z = \frac{x - \mu}{\sigma} \quad (3.8)$$

⁷`patch + np.random.normal(loc = 0, scale =  $\sigma_{patch_{aug}}$ )`

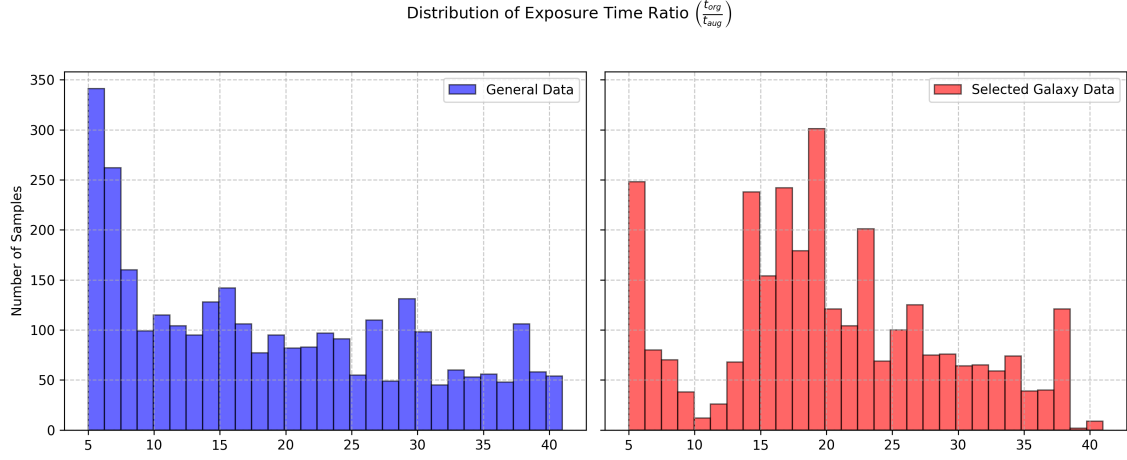


Figure 3.4: Distribution of exposure time ratios for the selected PI dataset (red, right) and general dataset (blue, left). The x-axis represents the exposure time ratio values, while the y-axis indicates the number of samples. The selected PI data shows a minimum at lower ratios around 10 compared to the broader distribution of the general dataset, presumably due to high number of images around 700-800 seconds exposure times, see the left diagram on 3.2.

In total, four networks with exactly the same architecture were trained:

1. on the selected PI dataset with z-scaling
2. on the selected PI dataset without scaling
3. on the general dataset with z-scaling
4. on the general dataset without scaling

The runtime for each of these with 1,000 epochs was around 17 hours on NVIDIA A40 GPU. At each 25. epoch, 32 patches were saved randomly from the validation data to track the network convergence.

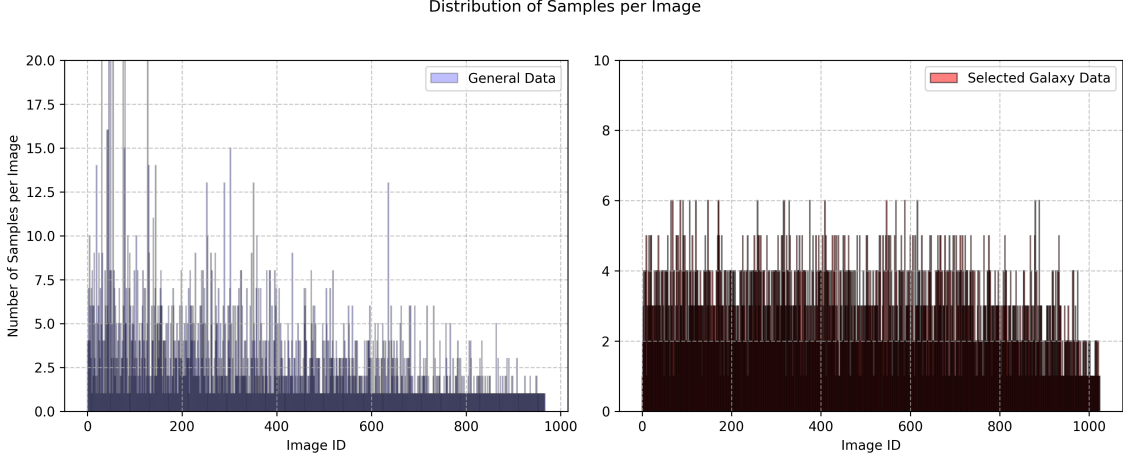


Figure 3.5: Distribution of samples per image ID for selected PI data (red, right) and general data (blue, left). The y-axis denotes the number of samples, and the x-axis corresponds to unique image identifiers. The selected surveys display a more homogeneous distribution with fewer samples per image ID in comparison to higher oscillations seen in the general dataset reflecting the more heterogeneous nature of the latter.

3.3 Detections

In order to detect and cross match sources between ground truth, noisy and reconstructed images; the python library **Source Extraction and Photometry (SEP)** was used. **SEP** builds on the widely used console-line **SExtractor** program for segmentation and analysis of astronomical images and aims at providing the basic functionality of the latter within a Python environment. It offers functions to estimate the background, detect objects and sum up fluxes around objects effectively calculating the Kron radius. In object detection task following parameters were used:

- **'thres'** : 2.25 (sigma threshold for object detection in the noisy and reconstructed images)
- **'thres'** : 2.0 (sigma threshold for object detection in the ground truth images)
- **'minarea'**: 8 (number of minimum neighboring pixels in the noisy and reconstructed images to be considered as part of an object)
- **'minarea'**: 6 (number of minimum neighboring pixels in the ground truth images to be considered as part of an object)
- **'deblend_nthresh'**: 32 (Number of thresholds for deblending)

- **'deblend_cont'**: 0.005
- **clean'**: True
- **'clean_param"** : 1

A more detailed information as to what these parameters do can be found in **SEP** documentation⁸.

⁸SEP Documentation

Chapter 4

Results

In this chapter, the results obtained from the trained model for estimating structural parameters and flux are presented. The model has been applied to the test data to extract key properties of galaxies, focusing on the measurement of fluxes and structural parameters such as the Kron radius which provide essential information about the size, morphology, and brightness of galaxies.

We find that the results obtained from the networks without any scaling underestimate the fluxes considerably. As this network does not produce astronomically accurate results, we will not show the outputs from it. Only the models that were trained using z-scaling will be shown. Among these, the models trained on the broad dataset and the dataset from the selected PIs show very similar trends. Therefore, only one of them, the one that was trained on the selected PIs will be discussed in the subsequent sections. Analogous diagram for the network trained on the general dataset can be found in the Appendix.

First, we will present some examples outputs from the network. Following that, we will compare the fluxes and the Kron radii of objects detected in the outputs from the network to those of the original images. Within each section, we will first discuss general trends between the reconstructed and noisy images relative the corresponding parameters in the ground truth images. A closer examinations using either the respective flux ratios or differences in Kron radii with respect to the ground truth AB magnitude will follow.

4.1 Visual Inspection

Figure 4.1 presents a mosaic of the network’s output simulated at different exposure ratios, denoted as $\gamma = T_{\text{org}}/T_{\text{simulated}}$, where T_{org} represents the actual exposure time and $T_{\text{simulated}}$ is the exposure time in the simulated (noisy) images. Z-scaling has been used for visual representation only. The image was taken from Extended Groth Strip (EGS), a field in Ursa Major constellation. The original image used for this example has a total exposure time of 1,805.87 seconds. Only a patch of 256 pixels x 256 pixels is shown.

The top row of the figure displays the ground truth image as the reference for all comparisons. In the second and fourth rows, the noisy images corresponding to the different exposure ratios are shown. These noisy images are what the network works with, and they simulate the real-world scenario where the data quality is degraded due to reduced exposure times. The third and fifth rows show the corresponding reconstructed images produced by the network, where the model attempts to recover the objects lost due to noise in the simulated images. As the exposure time ratio increases, the noisy images show a loss in the clarity and detection of objects. However, the network’s ability to recover the original objects from these degraded images is evident in the reconstructed images. The network effectively sharpens and refines the features that were initially obscured by noise, improving the overall clarity of the objects, even under reduced exposure conditions as high as $\gamma = 41$.

Figure 4.2 shows the same images with over-plotted ellipses of the objects using **SEP**. This figure highlights the specific detections in the ground truth as well as in the reconstructed images. Red ellipses correspond to mutual detections while blue ellipses illustrate detections found only in the ground truth. Green ellipses, on the other hand, display the detections found only in the reconstructed images. Some of these detections correspond to real objects in the ground truth image that were too faint to be captured by the **SEP** parameters even though the parameters used in the detection of noisy and reconstructed images were more conservative than those in the ground truth images (8 to 6 minpixel area and 2.25 to 2σ deviation). This can be attributed to the network’s ability to recover faint or partially obscured objects. Additionally, on closer inspection, we notice that there are some size differences between the objects in the noisy and reconstructed images. The comparison between the noisy and reconstructed detections highlights the network’s ability to not only recover objects that were missed but also its capacity increase the sharpness of the output image which can result in larger areas above the detection threshold for some sources.

As a counter-example of the well-behaved network outputs in the previous two figures, in Figures 4.3 and 4.4 we illustrate the network’s output simulated at different exposure ratios as well the detected objects in a case where the recovered imaged depart from the original. This image also originates from the Great Observatories Origins Deep Survey with an actual exposure time of 1,205.86 seconds.

Despite the relatively comparable exposure times with the previous example (the former had around 1,800 seconds), an interesting behavior is observed in the latter: the network *hypercorrects* by increasing the signal-to-noise ratio (SNR) beyond its actual level. This hypercorrection effect becomes more pronounced with higher exposure ratios, leading to reconstructed images that appear unnaturally enhanced compared to the ground truth in exposure time ratios above $\gamma = 8$ which should mimic the noise levels at an exposure time of 150 seconds. This over-correction may result in overly sharpened features or even the amplification of background noise, which is clearly visible in some regions of the reconstructed images.

Figure 4.4 complements this analysis by displaying the **SEP** detections on the individual

reconstructed images from Figure 4.3. In these images, small distortions and irregularities can be observed in the ground truth around the locations of many objects that are detected only in the reconstructed images (marked in green). These green ellipses suggest that while some of these detections correspond to genuine, faint objects recovered by the network through enhancement, others may simply be artifacts of enhanced noise. Furthermore, the background in these reconstructed images is not uniformly homogeneous and displays subtle patterns, which might be attributed to the network’s overcompensation during reconstruction.

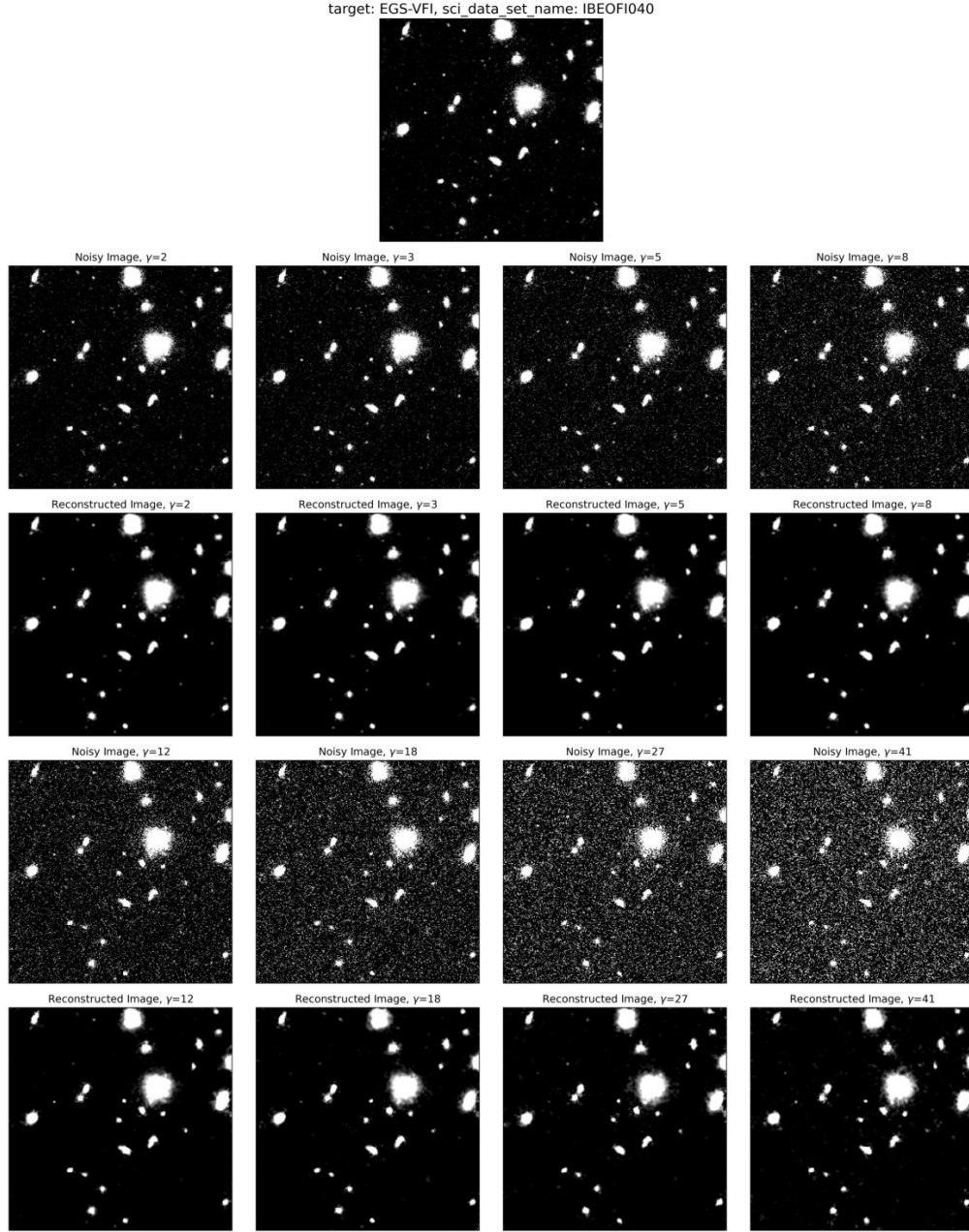


Figure 4.1: Mosaic of Networks’s Output simulated at different exposure ratios γ . The first row demonstrates the ground truth image. The 2nd and the 4th rows illustrate the noisy images the network saw at given γ and the images at the 3rd and the 5th rows display the corresponding reconstructed images. From the Extended Groth Strip, original exposure time: 1,805.87 seconds.

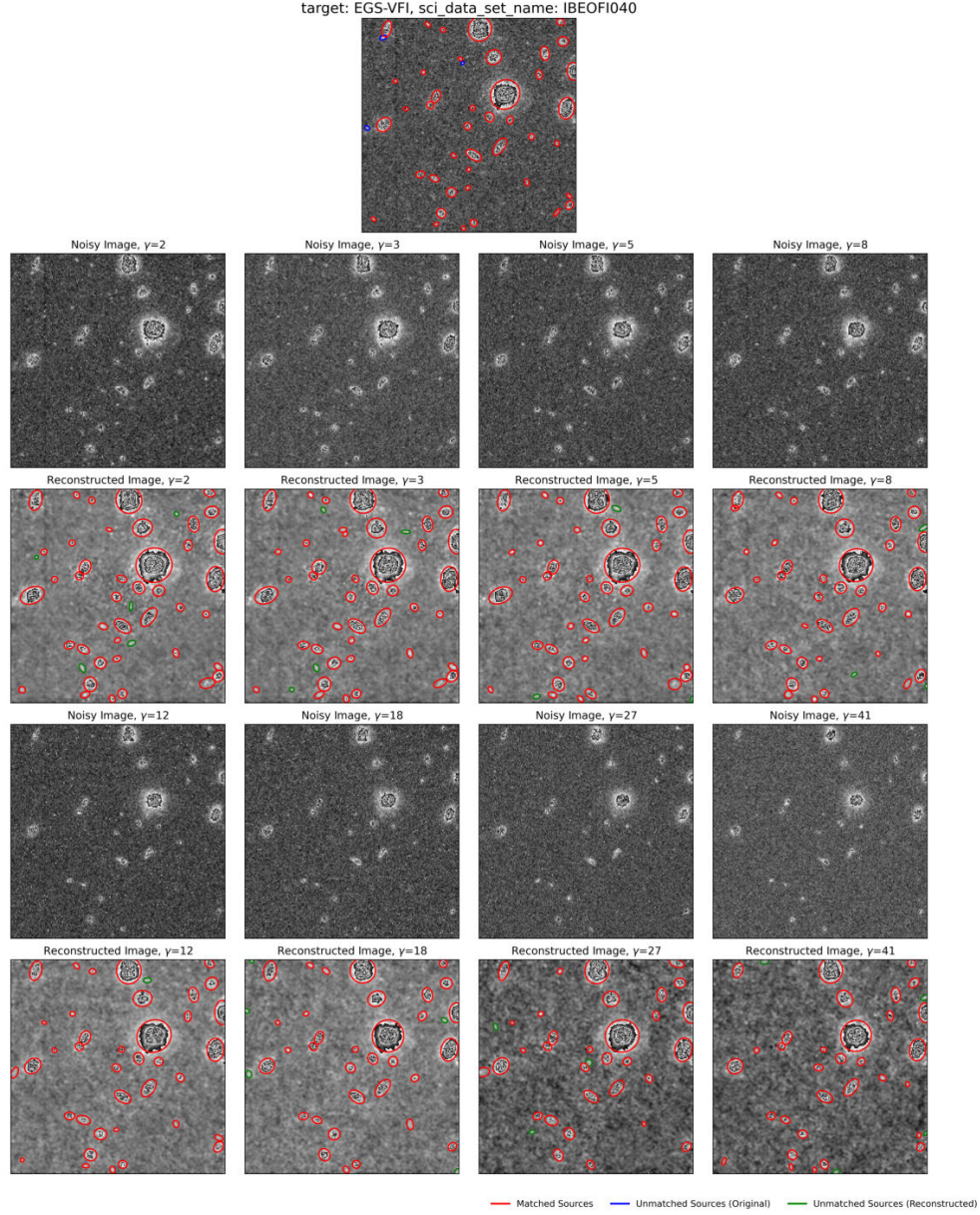


Figure 4.2: SEP detections on the individual reconstructed images from 4.1. Some of the detections only found in the reconstructed images (green ellipses) correspond to genuine objects found in the ground truth image that were too faint to be captured by SEP. This could have been solved by providing even smaller thresholds for detections in the ground truth image. The size differences between objects in the noisy and the reconstructed images is clearly visible. From the Extended Groth Strip, original exposure time: 1,805.87 seconds.

target: GOODS-N-ORI180-V10, sci_data_set_name: IBOEAL020

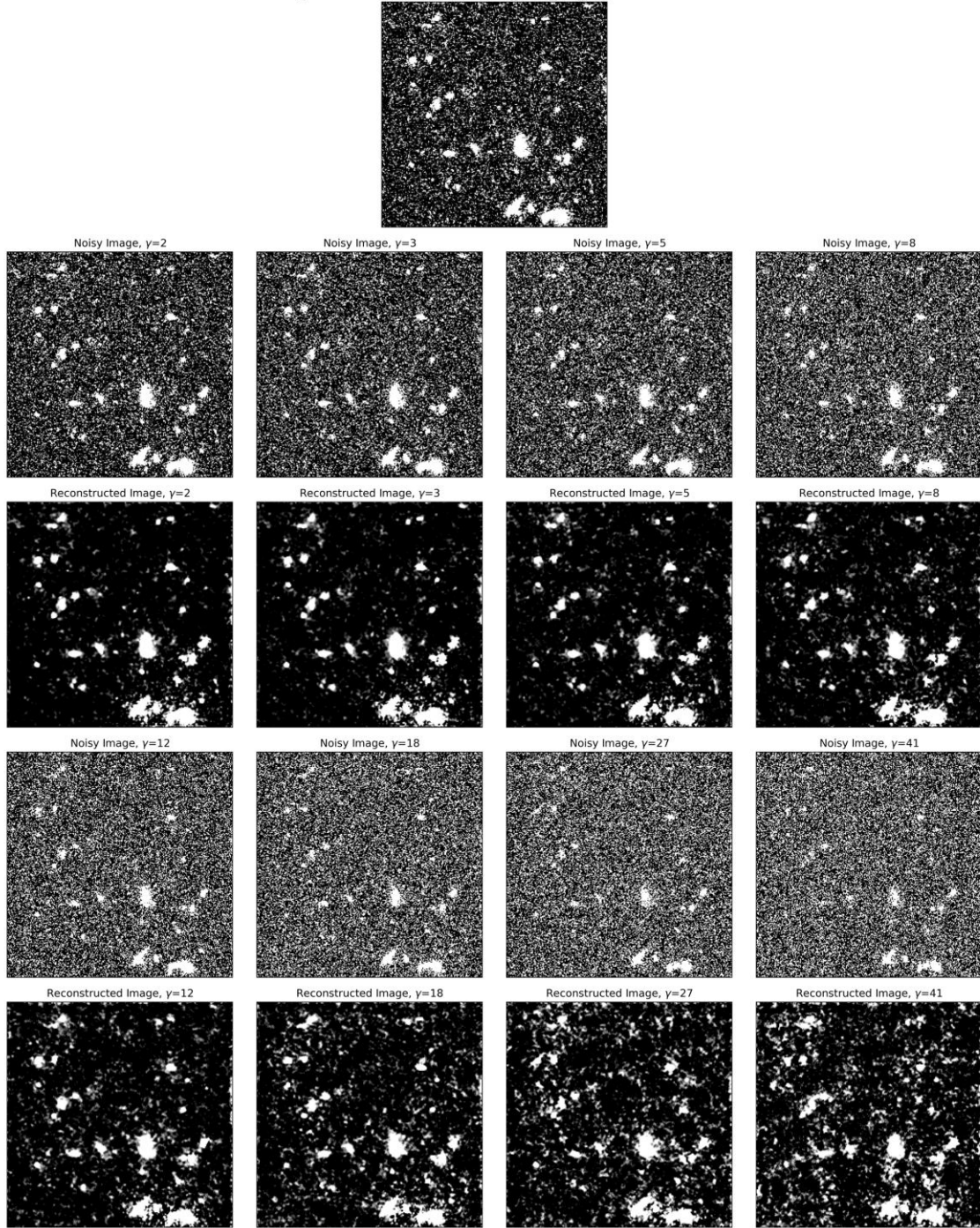


Figure 4.3: The network can hypercorrect in some cases and increase the SNR than its actual amount. This increases with exposure ratio γ . More information about individual detections are in 4.4. From the Great Observatories Origins Deep Survey. Actual Exposure Time: 1,205.86

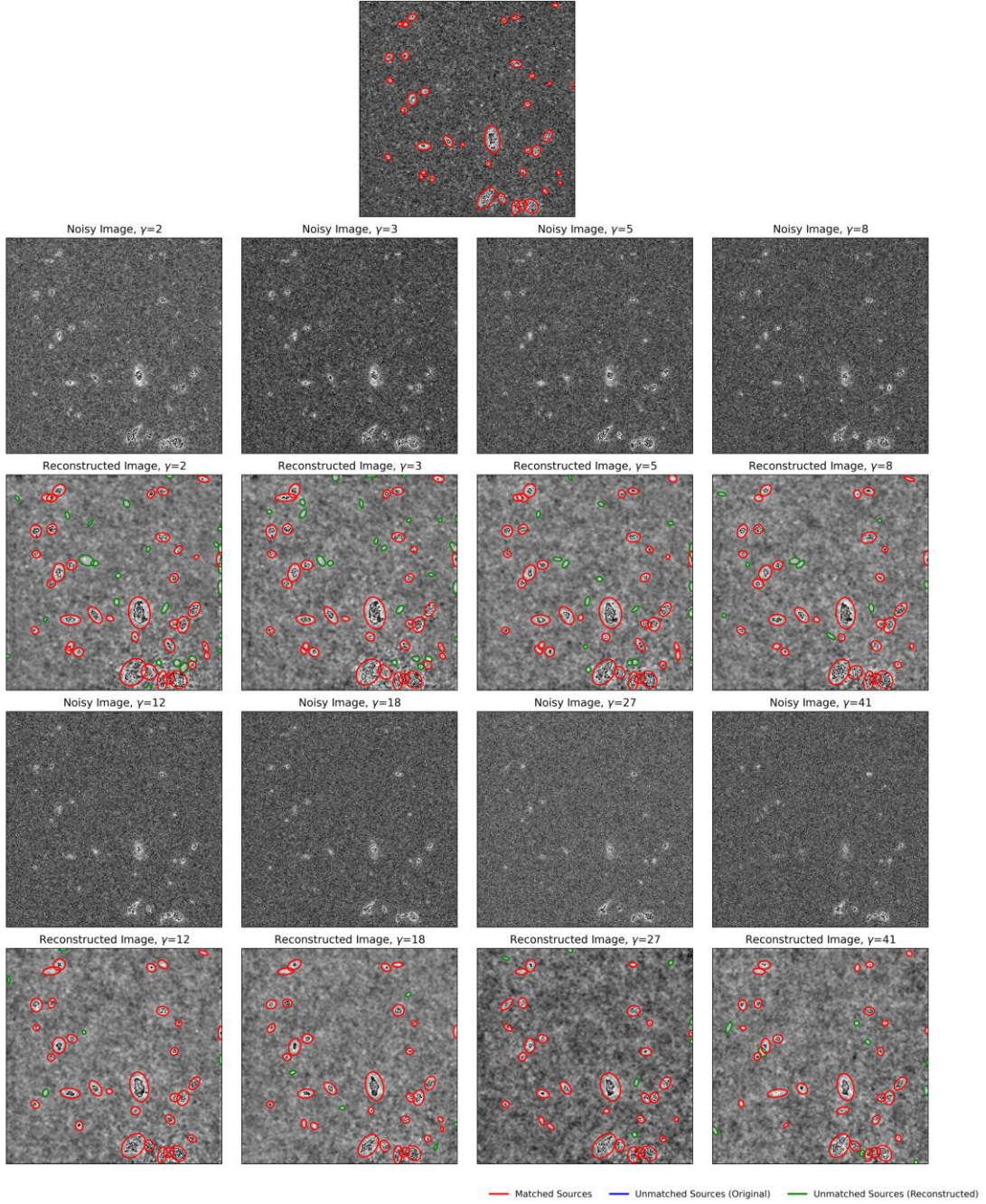


Figure 4.4: SEP detections on the the individual reconstructed images from 4.3. From the Great Observatories Origins Deep Survey. One can find some small distortions and irregularities in the ground image around the locations of many of the objects that are only detected in the reconstructed images (green). Some of this are presumably faint objects thorough enhancement while others are probably enhanced noise. The background seems to be not uniformly homogeneous and show some patterns. From the Great Observatories Origins Deep Survey. Actual Exposure Time: 1,205.86

4.2 Quantitative Metrics

We use a number of quantitative metrics to assess the performance of the network. These include:

- **True Positives (TP)**: is the number of objects detected both in the reconstructed as well as in the ground truth images.
- **False Positives (FP)**: is the number of objects detected only in the reconstructed images. These might contain sources that were too faint in the ground truth images as well as artifacts of the network that do not correspond to real objects.
- **False Negatives (FN)**: the number of objects in the ground truth images that could not be detected in the reconstructed images.
- **Precision**: measures how many of the predicted objects correspond to real objects detected in the ground truth images and is given by:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.1)$$

- **Recall**: also known as the *sensitivity* and measures how many of the actual objects in the ground truth images were correctly predicted and is given by:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4.2)$$

- **F-measure**: also known as the *F-score*. It is the harmonic mean of precision and recall, providing a balance between the two. A high F1-score indicates a good trade-off between precision and recall. It is given by:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.3)$$

- **Peak Signal-to-Noise-Ratio (pSNR)**: It is given in decibels (dB) and measures the quality of the reconstructed image compared to the original:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{L^2}{\text{MSE}} \right) \quad (4.4)$$

where L is the maximum pixel value in the original image.

- **Intersection of Union (IoU)**: measures the overlap between the intersection area of the detected objects between the ground truth and reconstructed images normalized by their union, given by:

$$\frac{I \cap K}{I \cup K} \quad (4.5)$$

where I is the area of the detected object in the ground truth and K the area of corresponding object in the reconstructed map. 0 means no intersection and poor performance, 1 on the other hand a perfect match.

- **Structural Similarity Index (SSIM)**: measures the similarity between two images from a human perception, considering luminance, contrast, and structure rather than just pixel differences and is given by:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (4.6)$$

where μ_x and μ_y are the mean pixel values, σ_x^2 and σ_y^2 their variances, σ_{xy} their covariance and C_1 and C_2 small constants to prevent division by zero.

Table 4.1: Summary of metrics for different exposure ratios.

exposure ratio	TP	FP	FN	Precision	Recall	F-measure	pSNR	SSIM	IoU
total	690420	140731	82387	0.83	0.89	0.86	72.43	0.98	0.69
2	139315	38906	4497	0.78	0.97	0.87	73.41	0.99	0.66
3	143138	34160	6779	0.81	0.95	0.87	72.96	0.99	0.67
5	128332	23834	11387	0.84	0.92	0.88	72.27	0.98	0.69
8	120536	18334	19458	0.87	0.86	0.86	71.68	0.98	0.71
12	65552	10266	15576	0.86	0.81	0.84	71.92	0.97	0.73
18	50717	7945	17061	0.86	0.75	0.80	71.69	0.96	0.73
27	3967	531	1999	0.88	0.66	0.76	72.84	0.96	0.71

Table 4.2 provides a detailed breakdown of these metrics across various exposure ratios, with $\gamma = T_{\text{org}}/T_{\text{simulated}}$. At the general level, we see that the total performance (aggregated over all exposure ratios) yields 690,420 true positives (TP), 140,731 false positives (FP), and 82,387 false negatives (FN) for AB magnitudes brighter than 25. These numbers result in an overall precision of 0.83 and recall of 0.89, with an F-measure of 0.86 which are measures of how well the network performs at detecting actual objects and which amount of the objects in the ground truth images are detected.

When examining the performance at specific exposure ratios, one can note that at $\gamma = 2$ and $\gamma = 3$, the recall values are exceptionally high (0.97 and 0.95, respectively), suggesting that nearly all true objects are detected under these conditions; however, this comes at the

cost of lower precision (0.78 and 0.81), implying a higher rate of false positives. However, many of these false positives actually correspond to objects that were not detected in the ground truth images which must be due to the parameters provided to **SEP**, as this was shown visually above. Therefore, these relatively low numbers cannot be reliable at these low exposure time ratios. As the exposure ratio increases to $\gamma = 5$ and beyond, precision improves up to 0.88 at $\gamma = 8$ while recall begins to drop, reaching as low as 0.66 at $\gamma = 27$. The F-measure remains relatively stable around 0.80–0.88. The constancy of this metric suggests that the model is robust across a wide range of exposure time ratios.

Quantifying at the image quality and flux metrics, the (pSNR) hovers around 71–73 dB across all exposure ratios, indicating consistent high reconstruction quality despite varying exposure settings.

The SSIM is uniformly high (around 0.96–0.99), demonstrating that the overall structure of the reconstructed images is well preserved relative to the ground truth.

Finally, the IoU, which measures the overlap between detected and true object regions, ranges from 0.66 to 0.73 with an average value 0.69. This says that 69% of the combined area of detections in the ground truth and reconstructed images overlap. The remaining 31% could either be part of the detections in the ground truth images that could not be found in the reconstructed images or part of the FP detections only found in the reconstructed map.

4.3 Flux Recovery

Flux is a fundamental quantity in astrophysics, representing the total energy received from an astronomical source per unit time. Accurate flux estimation is crucial for characterizing galaxies, as it directly relates to their luminosity and other physical properties. In this section, the results of flux estimation obtained using the trained network are presented. In the first section, the general trends will be discussed while the subsequent chapters will concentrate on individual trends across the detections captured in both reconstructed as well as noisy images.

4.3.1 Flux vs. Flux

First we compare the fluxes of the detections in the reconstructed and noisy images with respect to the corresponding fluxes in the ground truth images, shown in Figure 4.5. The points plotted are across the full set of test images, aggregated by noise ratio (γ , shown above each panel). The red dashed lines show the one-to-one lines.

The number of detections, N (shown above each panel), in the reconstructed images are almost twice as high relative to the noisy image, with the exception of $\gamma = 18$ and $\gamma = 27$ where they are tripled as all detections are considered without any previous filtering. Detections both in the reconstructed and noisy images cluster around the one-to-one line and the dispersion around this line increases towards fainter fluxes in both cases. This is expected as the error in flux estimation increases as objects become fainter.

Two general trends can be observed in Figure 4.5. The first one is that the fluxes in the noisy images are consistently underestimated. This should be of no surprise as the added noise results in decreased SNR for objects in the noisy images. Fewer pixels are associated with the objects as more pixels are lost to noise, effectively decreasing the area of the object leading to lower flux. In contrast, fluxes in the reconstructed images are consistently overestimated which indicates that the model sharpens the objects above the level of the original image.

Additionally, objects are detected at smaller fluxes in the reconstructed images than in the noisy images which is illustrated in the flux curves extending to $10^{-27} \text{ergs}^{-1} \text{cm}^{-2} \text{\AA}^{-1}$ in the reconstructed images while they stop at around $10^{-26.5}$ for γ values up till $\gamma = 5$ and at around 10^{-26} for higher γ values. This is most easily seen in the corresponding plots for $\gamma = 18$ and $\gamma = 27$. However, the dispersion of the said fluxes increases with the increase in range so that this higher coverage does not translate directly to an overall precision. Although the model consistently overestimates the fluxes up to this range, it starts underestimating them more than it overestimates them below the said range.

4.3.2 Flux Ratio vs. AB Magnitude

To better quantify the trend in flux recovery, in Figure 4.6 we divide the fluxes in the reconstructed and the noisy images by the fluxes measured in the ground truth images. On the x axis here we convert the ground truth fluxes to AB magnitudes as this is more intuitive and provides a clearer representation of deviations and trends across different brightness levels.

The first striking observation is the inverse shapes of the distributions seen between the flux ratios in the reconstructed and noisy images. This is in line with the findings of 4.5. The fluxes in the noisy images are underestimated due to a reduced SNR while the fluxes in the reconstructed images are overestimated due to an hyper-increased SNR. While in both cases, the points scatter fairly close to the one-to-one line at bright magnitudes, the dispersion seen in the reconstructed images increases rapidly starting from an AB magnitude of 25 which is already evident in $\gamma = 3$ but becomes more evident as the exposure time ratio increases to 12 and more. Nonetheless, the color intensity seen in the detections from reconstructed images is higher and more tightly concentrated around the 1 value in comparison to the corresponding diagrams in the detections seen in the noisy images which follow a less steep color gradient. This indicates that the network can estimate the fluxes closer to the ground truth than what would only be possible through noisy images despite the overestimation. Furthermore, the deviations are significantly lower around 22-24 AB magnitude range where the majority of the detections are below 0.25. This difference increases with the AB magnitude where it reaches 0.5 after 24 AB magnitude. This is, at least, visually seems to be where the majority of the detections are located.

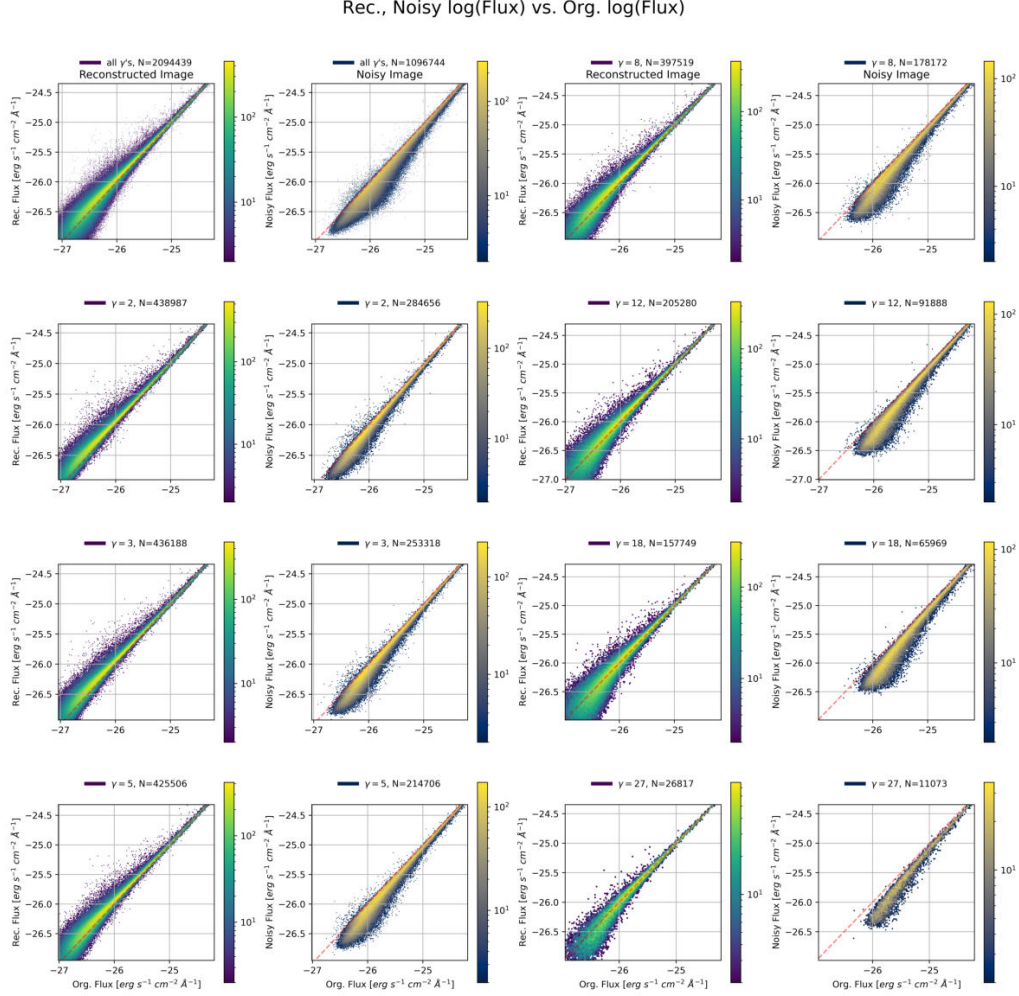


Figure 4.5: Figure shows the logarithm of flux for detections in reconstructed and noisy images compared to ground truth fluxes. The red dashed lines indicate the one-to-one relation. The number of detections in reconstructed images is nearly double that of noisy images, increasing to a threefold difference at $\gamma = 18$ and $\gamma = 27$. Detections cluster around the one-to-one line, with dispersion increasing at lower fluxes due to higher estimation errors for fainter objects. Fluxes in noisy images are consistently underestimated due to reduced SNR, while reconstructed images tend to overestimate fluxes, suggesting the model sharpens objects more than necessary. Additionally, reconstructed images detect fainter objects, extending flux curves to $10^{-27} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ Å}^{-1}$, whereas noisy images are limited to $10^{-26.5}$ for $\gamma \leq 5$ and 10^{-26} for higher γ . However, this increased range comes with higher dispersion, and at very low fluxes, the model transitions from overestimation to underestimation.

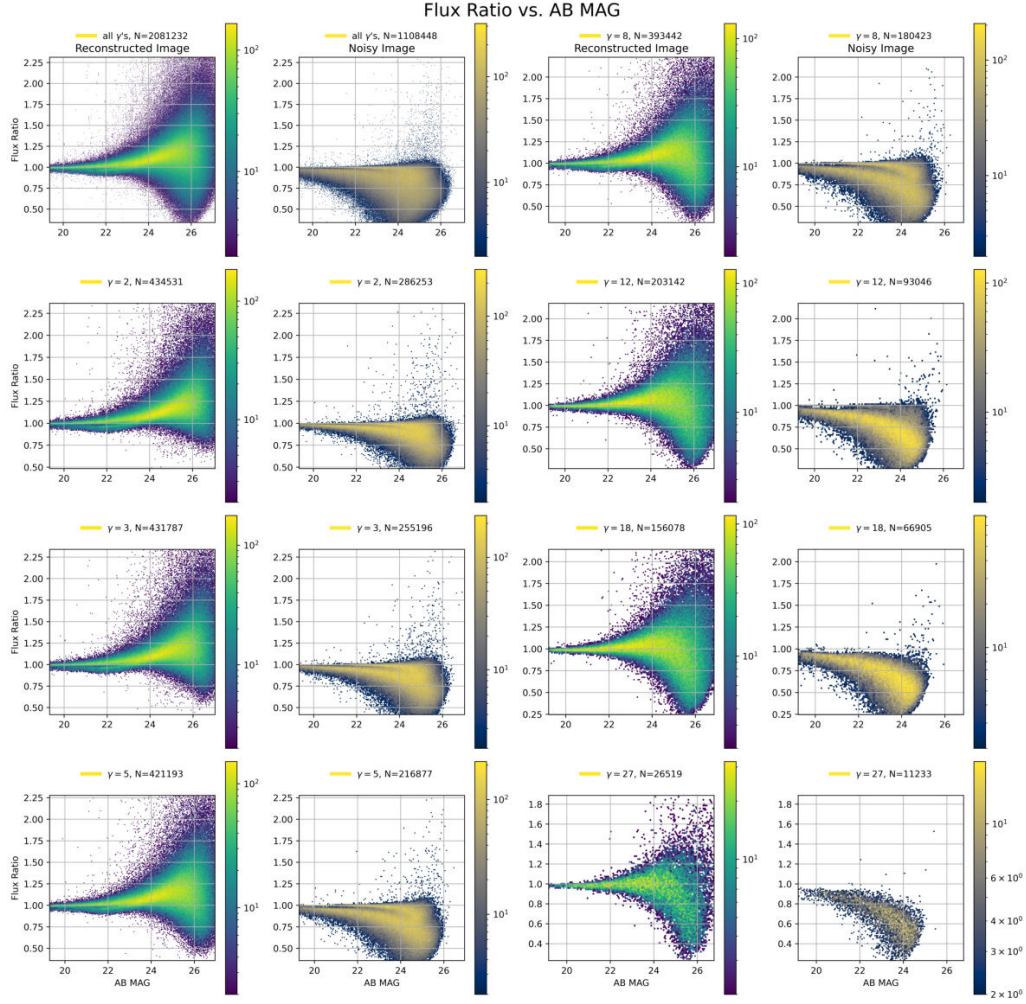


Figure 4.6: The flux ratios in reconstructed and noisy images show inverse distributions. Noisy images underestimate fluxes due to reduced SNR, while reconstructed images overestimate fluxes with an increased SNR. The dispersion in reconstructed images increases rapidly from an AB magnitude of 25, becoming more evident with higher exposure ratios. The color intensity in reconstructed images is higher around 1, indicating more accurate flux estimation compared to noisy images. Deviations are smaller for 22-24 AB magnitude, increasing to 0.5 for higher magnitudes.

4.3.3 Flux Ratio vs. AB Magnitude Below 25

The previous section has shown the limit of the model for objects fainter than an AB Magnitude of 25 AB. The errors start increasing rapidly, while the model starts underestimating the fluxes as much as it overestimates them. In addition, there were quantitative differences in the distribution of errors most easily seen between 22 and 24 AB magnitudes. Therefore, 25 AB magnitude can be interpreted as the upper limit of the model. Detections above this value will be filtered out in the successive plots to be able to study the performance of the network more in detail.

Figure 4.7 demonstrate the results. On a first notice, one sees that the dispersions became more regular due to the applied cut. The color intensities follow a similar path as already discussed above. However, one notices that the color gradient becomes less steep starting from $\gamma = 8$ in the reconstructed samples. This indicates a trend toward higher errors with increasing exposure time ratio. While the network tends to overestimation, it can also underestimate in the 20-22 AB Magnitude leading effectively to no bias within this range and a balanced performance. However, the overestimation increases in all diagrams starting from 22 AB. Furthermore, the center of the distribution shifts from around 1 towards 1.2 with the said increase. This can be deducted from all plots by following the trajectory of yellow pixels, however it is most easily recognizable at higher exposure time ratios such as $\gamma = 12$ where the yellow pixels in the reconstructed images are located closer to 1.25 than to 1 as the AB magnitude increases. A similar and more pronounced trend is also observed in the detections from noisy images albeit in the inverse direction. The deflection could be as high as 0.5 difference or more. In addition, the color gradient is significantly less pronounced which points to higher errors. This could be a possible explanation why the model tends to overestimate more at higher AB magnitude values as it might try to compensate the higher overall difference in the flux between noisy images and the ground truth by enhancing the objects at around these values than elsewhere.

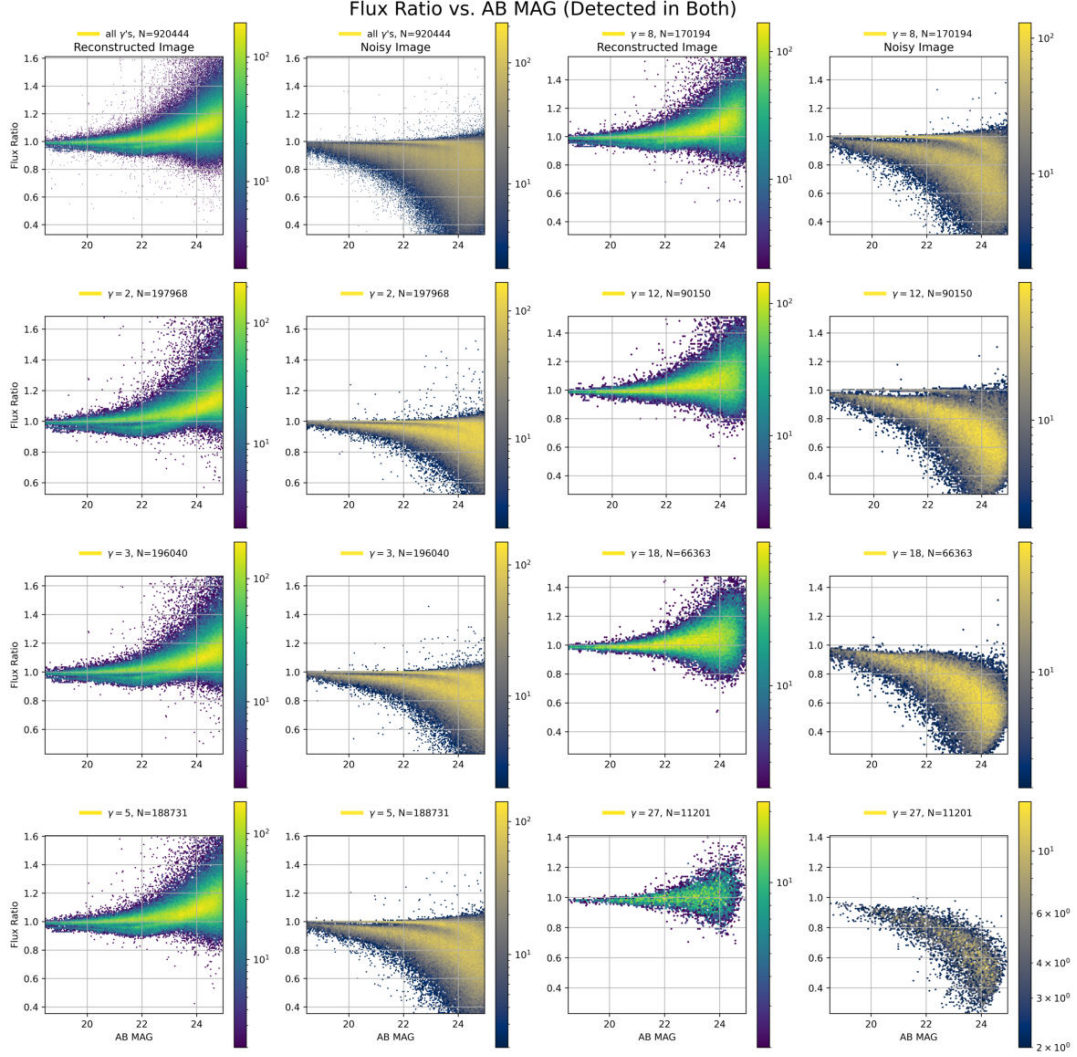


Figure 4.7: Figure shows the flux ratios for detections below 25 AB magnitude, where only objects detected in both reconstructed and original images are considered. The dispersions become more regular after the applied cut and increase with higher AB Magnitude. The color gradient in the reconstructed images becomes less steep starting from $\gamma = 8$, indicating higher errors with increasing exposure ratio. The distribution center shifts from around 1 to 1.2 as AB magnitude increases, with this trend most noticeable at $\gamma = 12$. At this same time, the model transitions from a more balanced more to higher overestimation. A similar but inverse trend is seen in noisy images, with deviations up to 0.5. The less pronounced color gradient in noisy images suggests higher errors, which may explain the model's tendency to overestimate flux at higher AB magnitudes, possibly compensating for larger flux differences between noisy images and ground truth.

4.3.4 Flux Ratio of Objects Found Only in the Reconstructed Images

Having shown the behavior of objects found both in the reconstructed and noisy images in the previous section, the properties of objects will be discussed in this section that were only found in the reconstructed images. This offers a more direct evaluation of the models performance as these objects are only found in the reconstructed images and fall within a range where the model is expected to perform relatively well.

Similar to the previous sections only detections below 25 AB magnitude will be considered which are shown in 4.8. Since there are no corresponding detections in noisy images, only a single color scheme has been used. The structure, nonetheless, is the same. The exposure time ratio γ increases from top to bottom and from left to right and the top left plot shows all detections regardless the exposure time ratio. In total, there were ca. 86,000 objects out of 1.1 million that were brighter than 25 AB magnitude and were only detected in the reconstructed images. This corresponds to $\approx 7.5\%$ of all detections below the said rate. The flux ratios of these objects are plotted against their brightness using the corresponding AB magnitude values in 4.8. One of first noticeable things is the range along x-axis which starts from AB MAG=23 which was 18 in 4.7. However, most of the detections are clustered between 24.5 and 25. This can be explained by the fact that objects that are brighter are less likely to be lost to the noise and hence more likely to be captured in both the noisy and reconstructed images.

Another interesting observation is that there are significantly less detections found at lower exposure time ratios of $\gamma = 2$ and $\gamma = 3$ that make up less than 6% of the detections found only in the reconstructed images. These are almost uniformly clustered around 25 AB Mag. This illustrates that the majority of objects found below this rate at the said γ values could be found in both images, albeit with higher errors for the detections in the noisy images as discussed in the previous section. The majority of the contributions for the objects only detected in the reconstructed images seem to come from exposure time ratios between $\gamma = 5$ and $\gamma = 18$ and from the AB magnitude range between 24.5 and 25. While the flux ratio can get as high as 2 if not 2.5, the likelihood decreases as one goes further away from 1 due to the exponential color scale.

Lastly, one sees that the center of concentration seems to lie halfway between 1.0 and 1.25 for all γ values, which is a clear sign of overestimation of the fluxes at around 12-13%.

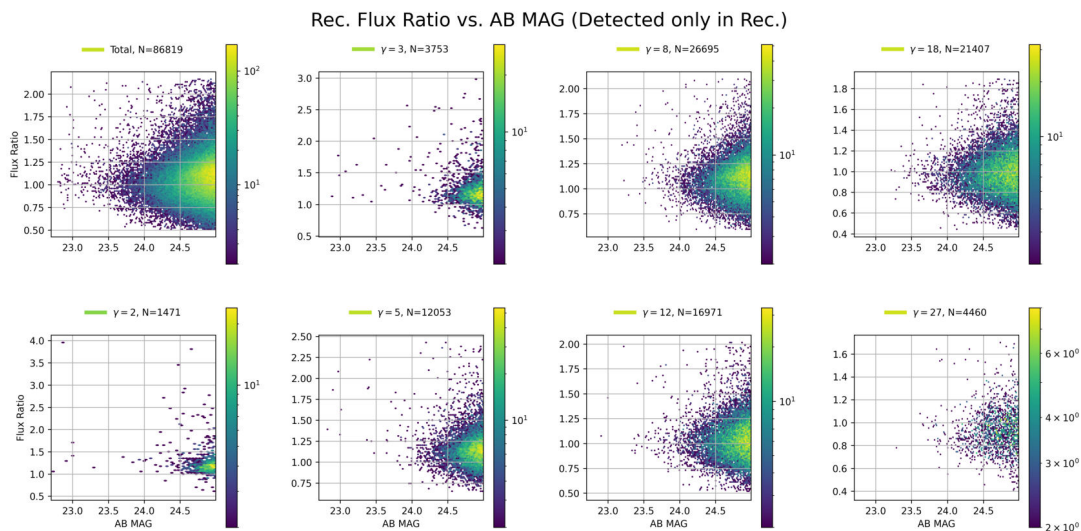


Figure 4.8: Figure shows the flux ratios of 86,000 detections with AB magnitudes below 25. Detections in the reconstructed images are mostly faint objects with AB magnitudes above 24, with flux ratios close to 1, though they can reach up to 2. Lower exposure time ratios ($\gamma = 2$ to $\gamma = 5$) show ratios that cluster tightly around 1, rarely exceeding 1.5. However, as these lower γ detections make up less than 10% of all detections, this does not significantly affect the overall trends. These objects likely retained more of their original fluxes at lower γ values, allowing for more reliable flux reconstruction. Moreover, the center of concentration lies halfway between 1.0-1.25 illustrating the overestimation of the model.

4.4 Estimation of Structural Parameters

Structural parameters describe the spatial distribution of light in a galaxy, providing crucial information about its morphology and size. One of the most widely used parameters is the *Kron radius* (Kron, 1980), which defines an adaptive aperture based on a galaxy’s light profile. Unlike fixed-aperture measurements, the Kron radius provides a flexible way to capture a significant fraction of a galaxy’s total flux while adapting to variations in its shape and brightness distribution.

The Kron radius is computed as a weighted mean radius, where the weighting function is the surface brightness profile of the galaxy. It is defined as:

$$R_K = \frac{\sum r I(r)}{\sum I(r)} \quad (4.7)$$

where $I(r)$ is the intensity at radius r . In practice, the flux within a circular aperture of **2.5 times the Kron radius** typically contains around 90% of the total flux of a galaxy. This makes it a robust parameter for defining photometric apertures, particularly in cases where galaxies exhibit varying sizes or irregular morphologies.

In this section, the results of structural parameter estimation using the trained model are presented. The model’s performance in determining the Kron radius is evaluated by examining the extracted Kron radii from detections in reconstructed and noisy images on the one hand versus the extracted Kron radii from the ground truth on the other hand. This will be followed by an error estimation in model using the difference in Kron Radius ΔK_{ron} with respect to brightness of the detected objects using AB magnitude to be able to examine possible trends that might differ between detections coming from reconstructed and noisy images with respect to their luminosity.

4.4.1 Kron Radius vs. Kron Radius

Figure 4.9 demonstrates the calculated Kron radii between the reconstructed (1st and 3rd columns), noisy (2nd and 4th columns) images along the y-axis and the ground truth images along the x-axis. They are ordered by their exposure time ratio ($\gamma = \frac{t_{org}}{t_{rec}}$). The exposure time ratio increases from top to bottom and each reconstructed image has a corresponding noise image directly right to it. The number of detections are displayed above each image and the densities are exponential. The red dashed line corresponds to the one-to-one line. Ideally, one would expect, all detections to be clustered tightly along this line. One of the first things one notices are the different N values between corresponding reconstructed (purple to green tones) and the noisy images (blue to yellow tones). In the total case (upper left), around 2.2 million objects were detected in the reconstructed images while almost the half this amount 1.1 million images could be detected in the noisy images. This 2-1 ratio persists among all different exposure time ratios all the way til $\gamma = 27$ where it reaches 3 to 1. Another easily noticeable thing is the concentration along one-to-one-line

for both the reconstructed and the noisy images as well as the relatively high dispersion along this line which can reach as high as $1''$ for detections in both images types. However, considering the exponential scale used in the plot lighter a single point along the one-to-one line can include as much as 1,000 detections while the darker shades along the edges could be below 10 detections. There seems to be an anomaly around $1.5''$ which is most easily seen in the accumulated detections captured in the noisy images from all different ratios (top row, 2^{nd} plot from the left). The Kron radii of many objects here are overestimated. This persists for noisy images for all covered exposure time ratios. Interestingly, such a skewness is not seen in detections of the model output which tends to be symmetrically centered along the one-to-one line. The model consistently corrects this anomaly which is most easily seen at higher γ values such around $\gamma = 12, 18$ or 27 . The center lines of detection in the noisy images deviate from the one-to-one line while those of the detections in the reconstructed images follow this. Lastly, the number of detected objects decreases while the minimum value of Kron radius increases at the same time, with increasing exposure time ratio. This is understandable as more and more objects are lost to the noise while only bigger ones persist.

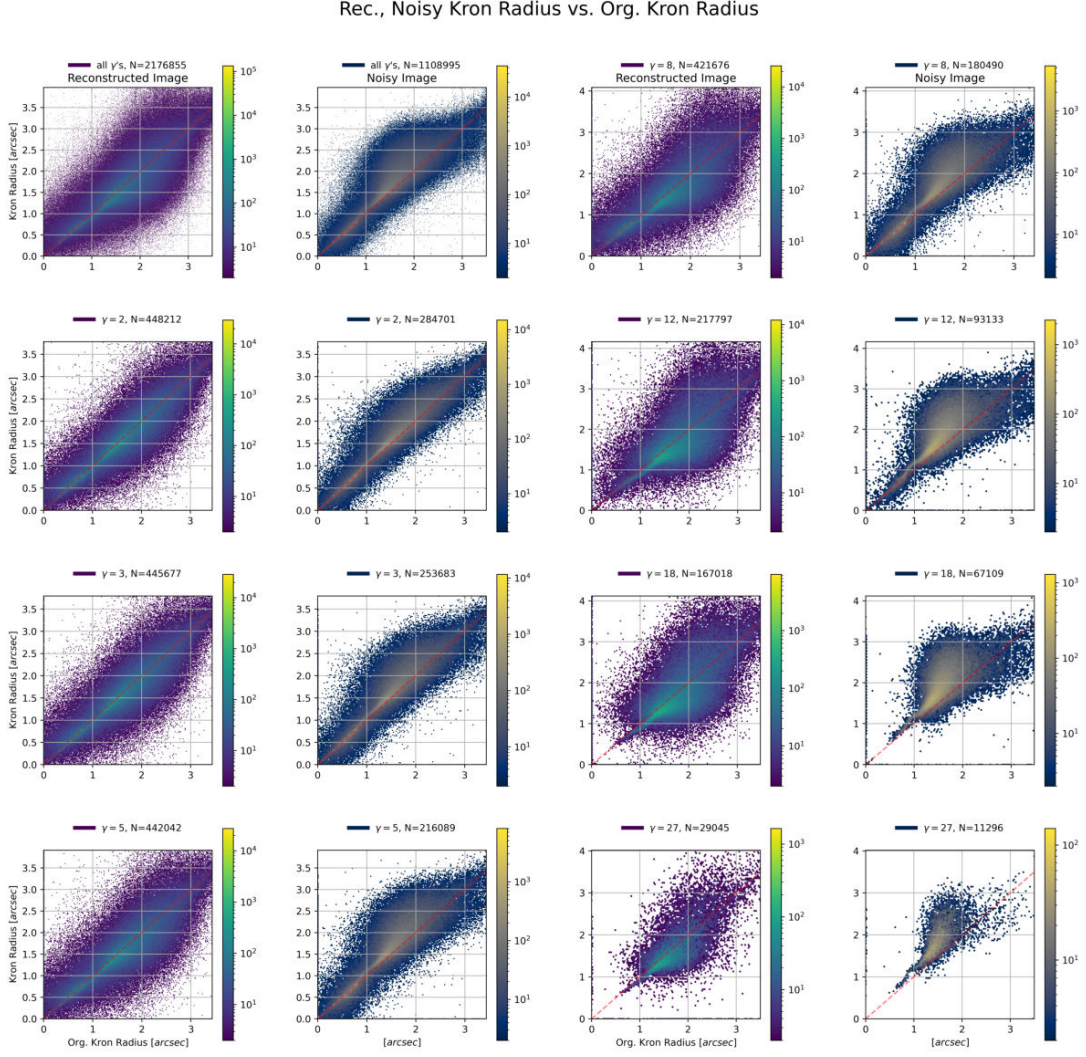


Figure 4.9: Comparison of the calculated Kron radius between reconstructed (1st and 3rd columns) and noisy (2nd and 4th columns) images along the y-axis, with ground truth images along the x-axis. The images are ordered by increasing exposure time ratio ($\gamma = \frac{t_{org}}{t_{rec}}$) from top to bottom. The red dashed line represents the one-to-one correspondence. While reconstructed images (purple to green) consistently detect more objects than noisy images (blue to yellow), a notable anomaly appears around 1.5" in the noisy images, where the Kron radius is overestimated. This discrepancy is corrected in the model outputs, which align more symmetrically with the one-to-one line, particularly at higher γ values. The number of detected objects decreases while their Kron radius increases with higher exposure time ratios, as smaller objects are lost to noise.

4.4.2 Δ Kron Radius vs. AB Magnitude

Plotting Kron radius versus Kron radius is useful to understand how the estimation of structural parameters behave across different sizes. However, objects of different sizes can be at any distance and the model might behave differently depending on the distance and the brightness of the detected objects. Therefore, it is informative to display the difference in the Kron radius of the detections in the reconstructed and noisy images from the detections in the ground truth ($\Delta\text{Kron Radius} = \frac{\text{Kron Radius}_{Rec,Noise}}{\text{Kron Radius}_{GT}}$) against the AB magnitude which is plotted in 4.10. The same plotting structure remains the same. The Purple/Green/Yellow color scheme corresponds to the reconstructed images and the Blue/Yellow scheme is kept for the noisy images. Exposure ratio increases from top to bottom. One of the first things one notices is the persistent cut of detections in the noisy images after an AB MAG value of 26 for smaller exposure time ratios and 25 for higher exposure time ratios starting from $\gamma = 8$. These objects are too faint to be detected in the noisy images which make up of almost the half of the detection in the reconstructed images. Generally, the difference in the Kron radius between detections increases in both type of images with increasing AB magnitude indicating that the error increases with what seems to be polynomial, possibly quadratic. While objects around 22 to 24 AB magnitude fall within an error of 0.5", this could reach 1.2" for objects above 25. Although, the distribution of the detections from both image types seem to follow similar trajectories, one sees on a more minute inspection that the errors of the detections in the noisy images are shifted above the zero line, whereas those of the detections from reconstructed images are centered around zero. Furthermore, the curve increases more steeply in the noisy images than in the detections from reconstructed images. This can be seen by looking at the color distribution around the edges, for instance by comparing the said distributions between 22 to 24 AB magnitude ranges at $\gamma = 5$. The color changes from green to purple rather abruptly in the reconstructed images, while it is more dispersed in the noisy images. Moreover, the color intensity around 0 seems higher in the reconstructed images, whereas this does not vary considerably in the noisy images where same coloring is retained till 1" error which can be more easily seen in exposure time ratios after $\gamma = 8$. 4.9

Rec., Noisy Δ Kron Radius vs. AB MAG

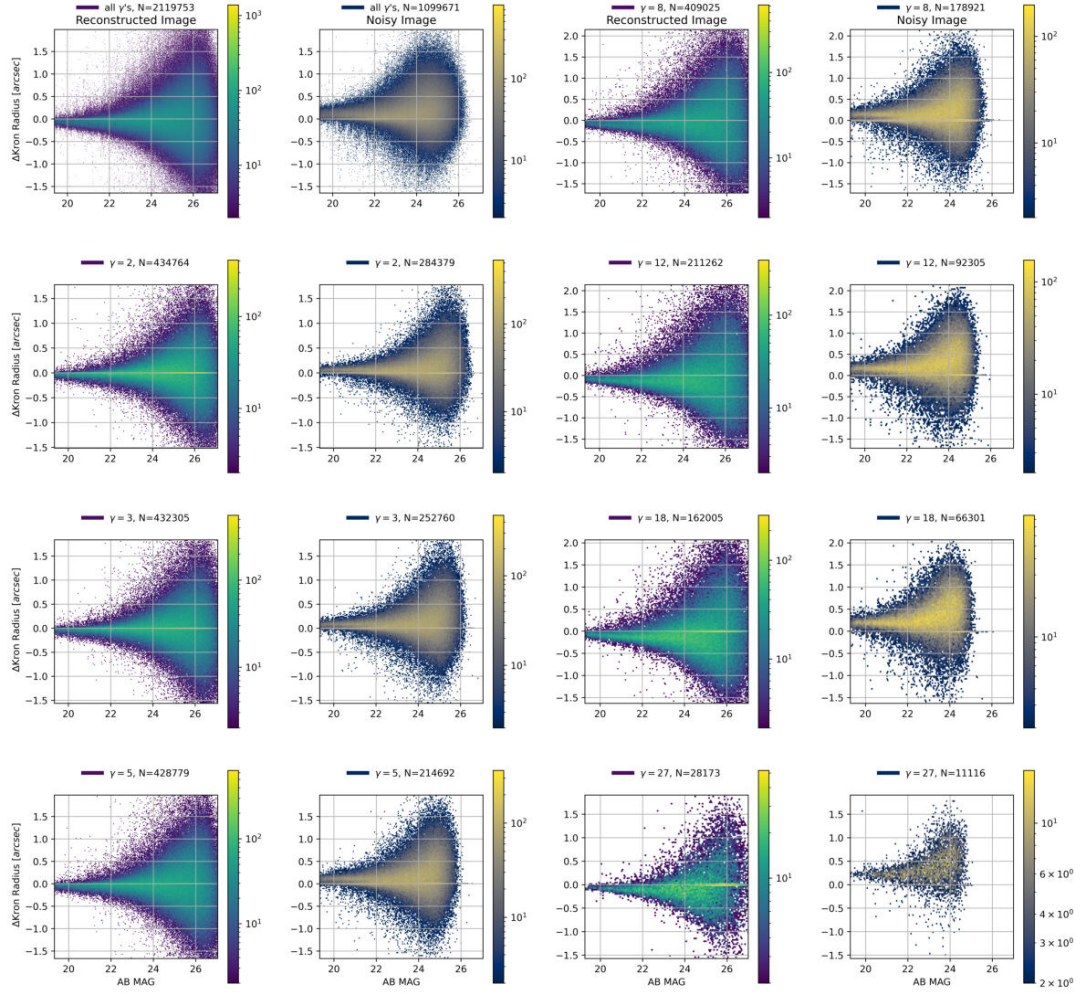


Figure 4.10: Difference in the Kron radius ($\Delta\text{Kron Radius} = \frac{\text{Kron Radius}_{\text{Rec,Noise}}}{\text{Kron Radius}_{\text{GT}}}$) between detections in the reconstructed (purple/green/yellow) and noisy (blue/yellow) images compared to ground truth, plotted against AB magnitude. The exposure time ratio increases from top to bottom. A clear cutoff appears in noisy images beyond AB magnitude 26 for lower exposure ratios and AB magnitude 25, indicating faint objects lost in noise. The error in Kron radius estimation increases with AB magnitude, reaching around 1.2" for the faintest objects. While both distributions follow similar trends, errors in noisy images are systematically biased above zero, whereas errors in reconstructed images are centered around zero. The steepness of the error increase is more pronounced in noisy images, as seen in the color distribution changes, particularly between AB magnitudes 22–24 at $\gamma = 5$. Reconstructed images show a more concentrated distribution around zero error, while noisy images exhibit a broader spread, especially at higher γ .

4.4.3 Kron Radius Below 25 AB Magnitude

The Figure 4.10 has shown that the error in Kron radius starts increasing rather fast after 25 AB Magnitude. This reiterates the need of filtering detections above 25 Magnitude which was already found while inspecting corresponding fluxes. We apply this filtering and show in this section the behaviour of objects captured in both images while the next will concentrate on only those found in the reconstructed images in a similar fashion to fluxes.

Figure 4.11 which is analogous to 4.9 demonstrates the Kron Radii of the objects in the reconstructed and noisy images on the one hand and the Kron Radii of the detections in the original images on the other. The number of detections is comparable for all exposure time ratios as the detections above 25 AB magnitude have been cut out. The dispersion around the one-to-one line seen in the detections from reconstructed images decreased considerably in contrast to those from noisy images whose shape has more or less stayed the same. The number of detections from noisy images has decreased ca. 200,000 in total while the same number has halved for reconstructed images which accounts for the narrowing down of the dispersion seen in this case. This being said, the trends discussed in 4.9 applies to the detections below 25 AB magnitude as well. The Kron radius of objects around 1.5'' is systematically overestimated in the detections from noisy images which are seen as bumps throughout individual plots and this overestimation causes the deviation of the central trajectory from the one-to-one line. Similarly to 4.9, this overestimation does not extend to reconstructed detections, which follow the one-to-one line more closely. This can most easily be discernible in higher exposure ratios, however it is also noticeable in lower exposure time ratios. Additionally, the color intensity in the reconstructed images is higher around the one-to-one line than the intensity for the same area in the noisy images which demonstrates that the network perform better than the noisy images and can recover Kron Radius more reliably than this can be extracted from noisy images.

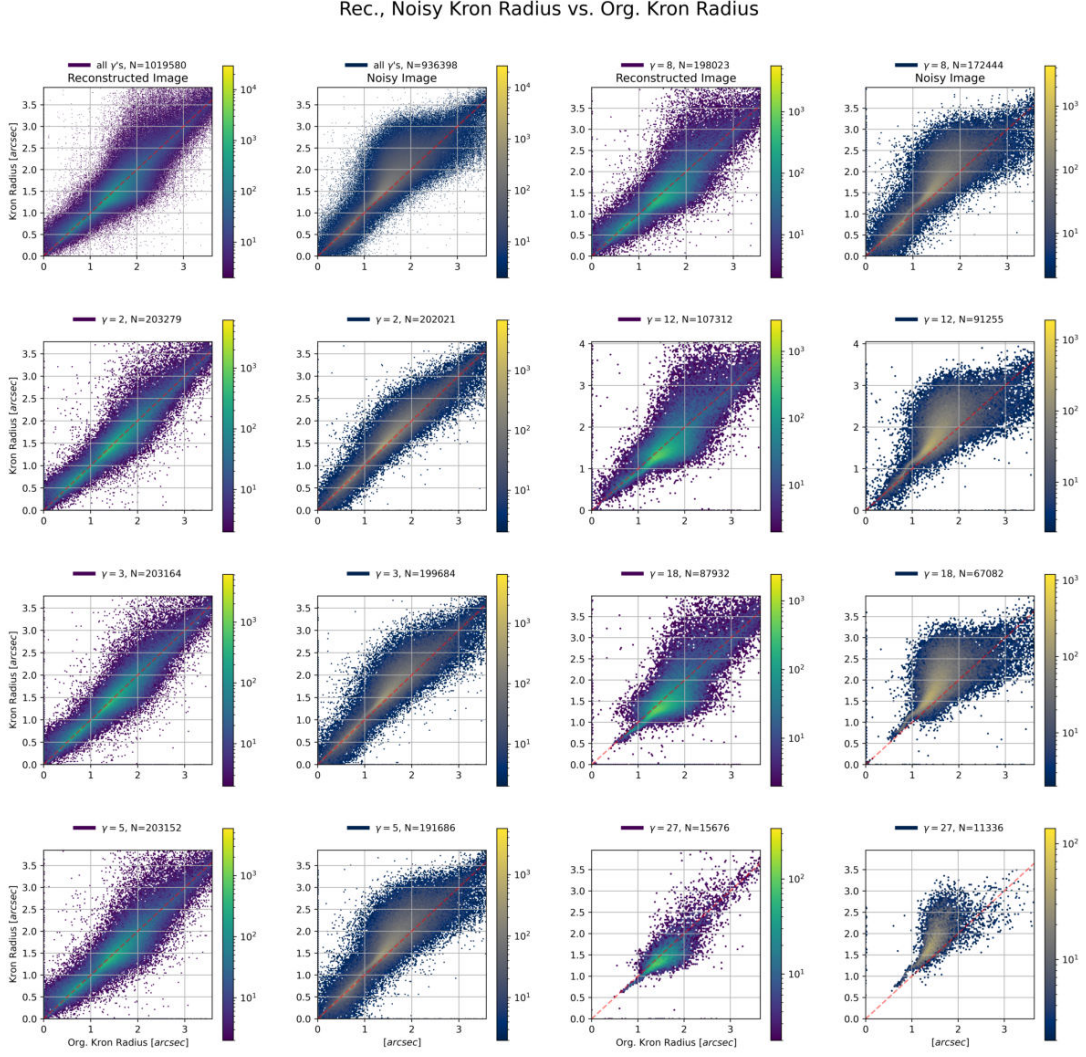


Figure 4.11: Similar to Figure 4.9, compares the Kron Radii in reconstructed and noisy images to the original images, excluding detections above 25 AB magnitude. The dispersion around the one-to-one line decreases for reconstructed images but remains similar for noisy images. A systematic overestimation around 1.5'' persists in noisy images, deviating from the one-to-one line, while reconstructed detections follow it more closely. Higher color intensity near the one-to-one line in reconstructed images indicates improved recovery of the Kron Radii compared to noisy images.

4.4.4 Δ Kron Radius vs. AB Magnitude Below 25

Similar to what was done in the previous example, the detections above 25 AB Magnitude have been cut to study the properties of the network more closely with respect to the differences in Kron Radius as a function of AB Magnitude. The examination of the Kron Radii of the objects which are only detected in the reconstructed images can offer insightful information about model's performance.

Figure 4.12 illustrates the properties of Δ Kron for the detections that were captured both in the reconstructed and noisy images. Therefore, N is exactly the same between both types of images. Although the shapes of the detections follow similar curves, high-intensity colors are noticeable in reconstructed detections as seen in exposure time ratios starting from $\gamma = 3$ than those found in the detections coming from noisy images, which tend to have more homogeneous low-intensity colors, indicative of higher errors found in the latter. The detections from reconstructed images are more tightly packed around the zero line. Therefore, the dispersion around the $\Delta R_K = 0$ line is narrower in the reconstructed images than in the noisy ones. Most detections in the reconstructed images are within Δ Kron $\sim 0.5''$ while the detections in the noisy images are more likely to be found within $1''$. This trend tends to be amplified as the exposure time ratio increases.

Furthermore, the dispersion of the detections from 0 in the reconstructed images increase consistently with higher AB Magnitude. While this is also true for the detections in the noisy images at lower exposure ratios, the shapes evolve into a more triangular shape and away from the zero line in higher exposure ratios towards lower Kron Radii than the actual amount such as at $\gamma = 12$, $\gamma = 18$ or $\gamma = 27$ which indicates that the Kron Radii are underestimated at these ratios in the noisy images. In contrast, these are more reliably estimated in the reconstructed images.

Rec., Noisy Δ Kron Radius vs. AB MAG (Detected in Both)

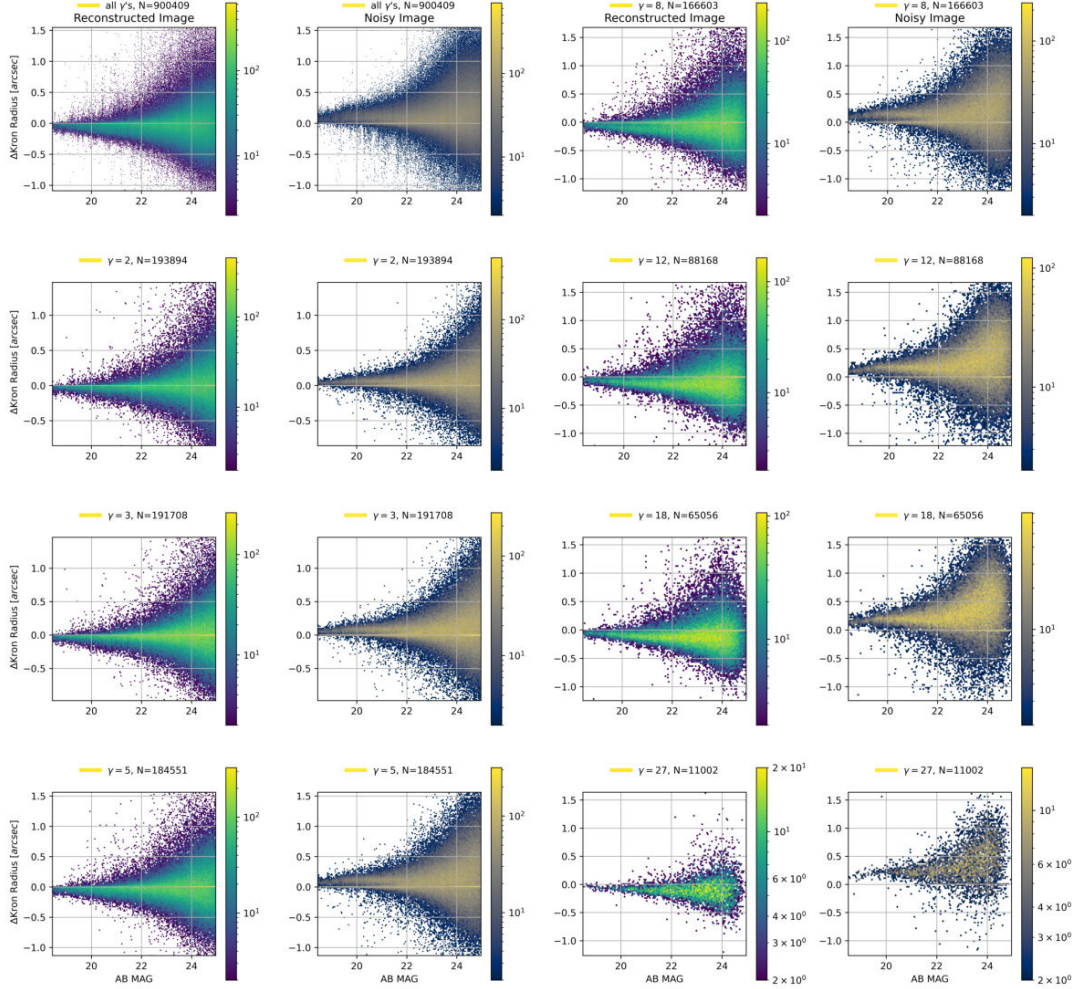


Figure 4.12: Figure shows the properties of Δ Kron for detections captured both in reconstructed and noisy images, with detections above 25 AB magnitude filtered out. The number of detections is identical between both image types. While the shapes of the detections follow similar curves, high-intensity colors are more prominent in reconstructed detections, especially at higher exposure time ratios ($\gamma \geq 3$), whereas noisy images display more homogeneous low-intensity colors, indicating higher errors. Detections in the reconstructed images are tightly packed around the zero line, with most within $0.5''$, while noisy images show detections within $1''$. This trend becomes more pronounced as the exposure time ratio increases. The dispersion in reconstructed images is consistent, while noisy images show a shift to a triangular shape at higher exposure ratios ($\gamma = 12, 18, 27$), suggesting underestimated Kron Radii in noisy images at these ratios.

4.4.5 Kron Radius of Objects Found Only in the Reconstructed Images

There are 83,000 detections that are found only in the reconstructed images, similar to what was given in 4.8. This was to expected as they cover the same objects. The shapes of the dispersions and the trends between 4.8 and 4.13 are also very similar. The main differences is the location of the concentration. While this is around 1 in the former with a shift to 1.12-1.13 illustrating the flux overestimation, the detections are concentrated around 0 in the latter without any bias. This is to be expected as the former pertains to a ratio between reconstructed and original fluxes, whereas the latter constitutes a difference in radii.

In a similar fashion to 4.8, Figure 4.13 also demonstrates that almost all the objects detected in the reconstructed images are faint objects above an AB magnitude of 24. While the ΔKron is closer to 0, it can go as high as 1" which is evident in the middle-intensity colors found in this range. However, this is rather unlikely. Lower exposure time ratios up to $\gamma = 5$ are an exception to this. For these lower γ values, ΔKron clusters more tightly around 0 and rarely goes to and above 0.5 which almost never happens for $\gamma = 2$ and $\gamma = 3$. However, as the detections in these lower exposure time ratios make up less than 10% of all the detections, this does not have a significant influence on the general behavior. Probably, objects still retained a higher degree of their original area at these lower γ values so that the network could reconstruct their Kron radii more reliably than it could do in higher exposure time ratios.

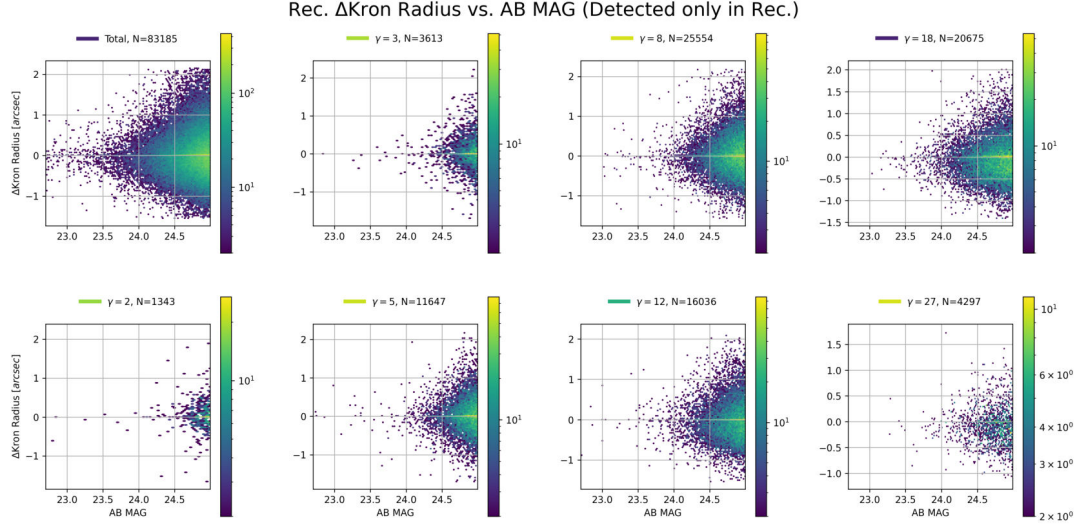


Figure 4.13: Figure shows detections below 25 AB magnitude, considering only objects found in reconstructed images, as there are no corresponding detections in noisy images. The exposure time ratio γ increases from top to bottom and left to right, with the top-left plot displaying all detections. Out of 1.1 million objects, approximately 83,000 ($\sim 7.5\%$) were detected only in reconstructed images. Notably, detections are clustered between 24.5 and 25 AB magnitude, as brighter objects are more likely to appear in both images. Lower exposure ratios ($\gamma = 2, 3$) contribute less than 6% of these detections, mainly around 25 AB magnitude, suggesting that most objects at these ratios are present in both images but with higher errors in noisy images. The majority of these detections come from $\gamma = 5$ to $\gamma = 18$. While individual errors can reach $1''$, their likelihood decreases exponentially with increasing deviation from zero. Additionally, the figure also demonstrates that no bias exists for Kron Radius in contrast to 4 where this is evident for Flux.

4.5 Summary of the Results

Lastly, we provide a summary of the results that have been introduced and analyzed visually in the previous sections. To this end, Table 4.5 presents descriptive statistics of the relative flux error (RFE) given by:

$$\text{RFE} = \frac{F_{\text{GT}} - F_{\text{rec}}}{F_{\text{GT}}} \quad (4.8)$$

where F_{GT} and F_{rec} are the flux of the ground truth and reconstructed images respectively. The said table also include statistics about ΔKron . Both quantities include the mean (μ), the standard deviation (σ) as well as the median. They are calculated across different

exposure time ratios (γ) and for all detections and those that are brighter than 25 AB Magnitude separately.

For flux errors, the standard deviation of the relative flux error remains relatively stable across different γ values, fluctuating around 17–21% with an average around 19.5% for sources brighter than AB MAG=25 and 32–41% with an average around 37% for all sources combined. Furthermore, the fluxes are overestimated which is observable in the minus sign that persists in all exposure time ratios. On average, they are around 15.91% overestimated for all sources and 8.8% for sources brighter than AB=25. However, the values fluctuate significantly depending on the exposure time ratio. For instance, fluxes are 10.85% overestimated at an exposure ratio of $\gamma = 2$ while the same error at $\gamma = 18$ is 3.4%. They are even 0.8% underestimated at $\gamma = 27$. A similar trend is also seen across all sources combined, however the rate could be up to two times higher than the one seen in brighter sources.

For Δ Kron, the standard deviation does not change considerably across different exposure time ratios and stays in the range (0.37" - 0.53") for objects brighter than AB=25. However, the standard deviation is quite large for fainter objects. The average here is 0.7". However, it increases significantly with exposure time ratio where it can reach 1.7" at $\gamma = 12$ or at $\gamma = 18$. In contrast to flux, there seems to be no bias towards over or underestimation.

To summarize the relative flux errors (RFE%) for all objects as well as for brighter objects are:

- $\text{RFE} = (-0.1591 \pm 0.3701)\%$
- $\text{RFE}^{AB \geq 25} = (-0.0880 \pm 0.1951)\%$

And Δ Kron:

- $\Delta\text{Kron} = (-0.0826 \pm 0.7034)''$
- $\Delta\text{Kron}^{AB \geq 25} = (0.0259 \pm 0.4126)''$

Table 4.2: Descriptive Statistics about the Errors Regarding Flux and Kron Radius Estimation For all 2 Million Detections as well as for those brighter than AB MAG=25

exp time ratio (γ)	$\sigma_{RFE}^{AB \geq 25}$	$\mu_{RFE}^{AB \geq 25}$	median $_{RFE}^{AB \geq 25}$	σ_{RFE}	μ_{RFE}	median $_{RFE}$	$\sigma_{\Delta\text{Kron}}^{AB \geq 25}$	$\mu_{\Delta\text{Kron}}^{AB \geq 25}$	median $_{\Delta\text{Kron}}^{AB \geq 25}$	$\sigma_{\Delta\text{Kron}}$	$\mu_{\Delta\text{Kron}}$	median $_{\Delta\text{Kron}}$
total	0.1951	-0.0880	-0.0597	0.3701	-0.1591	-0.0988	0.4126	0.0259	0.0079	0.7034	-0.0826	-0.0135
2	0.1958	-0.1085	-0.0795	0.3954	-0.2239	-0.1494	0.3714	0.0202	0.0402	1.1508	-0.0107	0.0055
3	0.2119	-0.1066	-0.0756	0.4118	-0.2065	-0.1330	0.3834	0.0223	0.0396	1.0625	-0.0208	0.0000
5	0.1976	-0.0975	-0.0651	0.3316	-0.1688	-0.1024	0.3970	0.0280	0.0389	1.0820	-0.0395	0.0000
8	0.1875	-0.0842	-0.0525	0.3299	-0.1276	-0.0711	0.4147	0.0285	0.0342	1.1767	-0.0698	0.0000
12	0.1700	-0.0612	-0.0330	0.3362	-0.0796	-0.0337	0.5290	0.0357	0.0300	1.7124	-0.0517	0.0000
18	0.1769	-0.0340	-0.0135	0.3516	-0.0361	-0.0055	0.4171	0.0347	0.0219	1.7001	-0.0598	0.0000
27	0.1715	0.0083	0.0111	0.3192	0.0292	0.0236	0.4126	0.0259	0.0079	0.7034	-0.0826	-0.0135

Chapter 5

Discussion

In this thesis, a 5 layered CNN network based on U-Net architecture as conceptualized by (Ronneberger et al., 2015) have been trained for the purpose enhancing the SNR of astronomical images, concentrating on extragalactic surveys using infrared images. The thesis was motivated mainly by the findings of Vojtekova et al. (2020) who trained a U-Net architecture on bright stars in the UV and the visible range of HST. (Vojtekova et al., 2020) report that overall 95.9% of stars with an average flux error of 2.26% were recovered. Additionally, the authors suggest that the recovered images had on average 1.63 times higher SNR than the noisy images provided to the network which is equivalent to stacking at least 3 images which may lead to a significant reduction of surveying time. While these findings are encouraging and motivate the use of U-Net architecture in astronomical denoising, it is not realistic to expect similar results from U-Net or similar GAN methods when applied to galaxies which are extended sources and are therefore morphologically quite different than stars.

Due to the faintness and the structural complexity galaxies possess, it is essential to compare the results of this thesis with studies that concentrate on the galaxies. Schawinski et al. (2017) trained a GAN on 4,550 images of nearby galaxies between $0.01 \leq z \leq 0.02$ from the Sloan Digital Survey. The network performs considerably well than a simple deconvolution method. While the said study includes visual evidence and metrics such as noise levels and pSNR, it does not contain metrics to quantify the recovery rate with respect to key parameters such as fluxes or structural parameters such as the estimation of Kron Radius and their change with brightness. While this is understandable from the perspective of Schawinski et al. (2017) as they concentrate on nearby galaxies at similar red shifts with similar structures, in this thesis we aim to delve further. The data considered in this thesis that originate from deep galactic surveys. In order to quantify the network's performance, different metrics have been put into use which include DL related ones such as Precision, Recall and F-Measure, image quality related metrics such as SSIM and pSNR as well as astronomy related metrics such as IoU , the ΔFlux or ΔKron . Additionally, flux

and Kron Radii were estimated with their respective ratios and differences as a function of AB magnitude to examine the network’s performance with respect to source brightness.

The precision is a measure which shows how many of the detected objects in the reconstructed images were actual sources and how many were due to network’s output or artifacts whereas the recall demonstrate the network’s ability to capture the amount of sources in the ground truth. F-measure is the harmonic mean of these two measures. The overall precision is 0.83 and recall 0.89, with an F-measure of 0.86. When examining the performance at specific exposure ratios, it can be seen that at exposure time ratios of $\gamma = 2, 3$ the recall values are exceptionally high (0.97 and 0.95, respectively), suggesting that nearly all true objects are detected under these conditions; however, this comes at the cost of lower precision (0.78 and 0.81), implying a higher rate of false positives. However, as demonstrated in the visual examples, the lower values at lower exposure time ratios cannot be directly interpreted as a failure of the model, as there are discernible objects by eye that were not captured in the ground truth images. This problem might be mitigated by applying different thresholds to the SEP for ground truth images on the one hand, and noisy as well as reconstructed images on the other. As the exposure ratio increases to $\gamma = 5$ and beyond, precision improves (up to 0.88 at $\gamma = 8$ while recall begins to drop, reaching as low as 0.66 at $\gamma = 27$). This trend indicates that, with increasing exposure ratio, the network becomes more conservative in its detections—fewer false positives occur, but some true objects are missed, likely due to the noise characteristics changing with exposure settings. However, the stable F-measure rates are an indication of robustness. Furthermore, they are also in line with the findings of (Vojtekova et al., 2020) for the same parameter.

Looking at the image quality and flux metrics, the peak Signal-to-Noise Ratio (pSNR) clusters around 71–73 dB across all exposure ratios, indicating consistent reconstruction quality despite varying exposure settings. These are quite high values and may be interpreted as part of the model’s tendency to overestimate the fluxes.

The Structural Similarity Index (SSIM) is a measure which assesses image quality based on perceived changes in structural information by humans, along with luminance and contrast. Essentially, SSIM provides a way to evaluate how natural reconstructed images look like. SSIM values are uniformly high (around 0.96–0.99) for all exposure time ratios, which shows that the overall structure of the reconstructed images is well preserved relative to the ground truth. In comparison, the same results by (Vojtekova et al., 2020) are placed around 0.5 which illustrate a higher deviation from a natural human perception. While such high rates should be considered with caution, the visual evidences provided in the previous section offer a justification for these high rates, since they look natural from a human perspective. Furthermore, the results obtained by (Zaazou et al., 2025) might provide some insights to these high values, as they trained GAN which they put to use to create mock galaxy images from DECaLS survey using different wavelengths including the ultraviolet, visible and the infrared and they consistently achieved similar SSIM values. Furthermore, these results are also in line with the findings of (Qi et al., 2022) who applied a modified U-Net architecture on 140 artificially corrupted images captured in UVIS channel

consisting mostly of stars, similar to Vojtekova et al. (2020).

Beside these metrics, the estimation of the flux and Kron Radius with their respective error estimations revealed some important insights into the model’s performance. It has been shown for instance, that half of the detections in the reconstructed images amounting to over 1 billion detections have no counterparts in the noisy images, and stem from very faint objects above an AB magnitude of 25. This also seems to be the range, where the errors for both Kron Radius and the flux start to increase rapidly. Above this magnitude the Kron Radius estimation might have an error up to $1.70''$ depending on the exposure time ratio with and the fluxes up to the 41% of the original flux depending on the exposure time ratio with average error σ values around $0.70''$ and 37% of the original flux respectively. Moreover, flux is 16% higher estimated in general. Due to these high uncertainties, the detections above 25 AB magnitude which was estimated to be upper limit of the network, have been filtered out. This led to the halving the amount of the detections. By doing that the mentioned quantities drop to $(0.03 \pm 0.41'')$ and $(-0.09 \pm 0.20) \%$ of the original flux as shown in 4.5.

Around 10% of the detections below 25 AB magnitude are only found in the reconstructed images which tended towards higher AB magnitude values in the range of 24 and 25. Following this, the fluxes and Kron Radii were first plotted against each other which were followed by the error estimations with respect to AB magnitude for objects detected both in the reconstructed and noisy images. The results show that the Kron Radii of the objects detected in the noisy data are consistently overestimated around $1.5''$ which leads to increasing deflections of Kron radii from the one-to-one line with increasing exposure time ratio between the actual duration time and the animated one. This might be due to object extraction of **SEP** and could be mitigated by a selection of better suited parameters. Such a deflection is, however not seen in the reconstructed images which were concentrated along one-to-one line. Furthermore, there seems to be no bias in the estimation of Kron Radius of detections in the reconstructed images even though the individual errors increase with AB magnitude. Moreover, color intensities are higher in the reconstructed images. Regarding Kron radius, the model seem to work best between 20 to 24 AB Mag where the errors are below $0.5''$. For the range between 20 to 22 AB magnitudes, the models reconstruct the size almost completely. These results are encouraging and speak for the use of U-Net for recovering structural parameters.

While the absence of bias in the Kron Radius is a good sign for the network, the estimation of the fluxes presents a different picture. The fluxes in the noisy images were consistently underestimated while those in the reconstructed images are consistently overestimated. While the former should be of no surprise due to the decreased SNR in the former resulting from the loss of boundary pixels to noise, the overestimation in the reconstructed images illustrates that the model overshoots and hypercorrects. This is evident in the negative RFE values seen 4.5. With an average relative error of around 20% despite the cut at 25 AB magnitude, the network cannot reliably reconstruct fluxes. As shown in Figure 4.7, the network transitions from minimal bias at AB magnitude 20–22 to a

systematic overestimation, approaching the midpoint between 1.0 and 1.25 as magnitude increases.

Two explanation for the high flux errors can be provided. The first is the steep decline in the primary trend where the majority of detections lie, as seen in the noisy-image detections with high exposure ratios in the aforementioned figure. This decline, which begins at AB magnitude 20 but remains relatively flat until 22 before dropping sharply, may have led the network to overcompensate the fluxes here more than in other areas. The second one is the distribution of the faint objects in the dataset. The second explanation is the dataset’s distribution of faint objects. Since most detected objects are quite faint (AB magnitude ≥ 24), the overall average flux error is skewed higher than it would be if the dataset were more evenly distributed, particularly with a larger proportion of objects in the AB magnitude 22–24 range. Despite this drawback, the reconstructed-image detections exhibit a tighter clustering than their noisy-image counterparts, supporting the model’s usefulness in improving flux recovery.

Chapter 6

Conclusion and Future Work

The aim of this thesis was to implement a U-Net architecture with the goal of training them on HST F160W images for denoising purposes. For this purpose over 1,623 images downloaded from the MAST database. A split of 0.7 was used to split the data into train and validation/test data. The network was run for 1,000 epochs. At each epoch, the network saw at least a (256 pixels x 256 pixels) patch of the images in the training dataset. Multiple networks with the same configuration were trained differing only in the dataset or scaling. All trained networks were tested on validation for the selected galaxy surveys. Results obtained from the networks with zscaling do not differ much from each other while the networks without any scalings perform poorly.

Results show that the network should not be used on objects that are fainter than 25 AB magnitude as the errors increase considerably. Visual evidence show that the network can reconstruct noisy images very well with an SSIM values of 0.99. The estimation of Kron radius and fluxes show that the shapes are reconstructed without any bias with a mean error of 0.41" while the fluxes are in average 9% overestimated with an error of 20%. However, this is highly dependent on brightness. These numbers apply after cutoff at 25 Magnitude. Further statistics regarding these numbers are summarized in 4.5.

Despite the relatively high uncertainties, the model could be a start and developed further. Even though, the results discussed here come from a subset the network has been trained on, it performs poorly on very faint objects. This should be considered in future works when training and applying U-Net on faint objects. Extending the network to reliably estimate fainter objects require further modifications. (Liu et al., 2025b) for instance use partial convolutions instead of full convolutions in their modified U-Net architecture which might help eliminating the effect of the dropped pixels during the dropout layer which were also used in our network. This can increase the effectiveness and efficiency improvement observed in practice. Another improvement could be by integrating attention mechanisms to the network. At the current state, the network has no way of knowing which parts of the image might be of interest and which parts are not, which covers the majority of the area

in astronomical images, as these are part of the background. This has been successfully implemented by Marini et al. (2024) where the authors integrated an attention mechanism to a regular U-Net architecture enabling it focus on regions of interest without further supervision. This has been achieved by adding these within skip connections between the contracting and expansive paths of the U-Net. The authors used this modified network to disentangle the stellar population in the central galaxy from the intrahalo light. Integrating an attention mechanism in our model to make the network focus on fainter objects covered in this work might help decrease the high uncertainties seen above 25 AB magnitude.

Lastly, the initial aim of this thesis was to apply the model on shallow 3-DASH images. While this has already been done on patch of images, this must be done on the whole image. Initial attempts resulted in artifacts around the edges where patches are overlaid. Averaging these had not solved the problem. Due to the tight time constraints, this had to be postponed. The immediate task will be tackling this problem.

Beyond the scope of the current work, U-Nets can be a promising avenue for image analysis in astronomy. As time is a valuable commodity for astronomical instruments, this approach can enhance the quality of data with no investment of additional time. Potential datasets for further exploration include the Euclid near-infrared imaging and, in the future, observations from the Roman Observatory.

Complementary Materials

The code used in this thesis is made public for the scientific reproducibility and can be found at [GitHub](#).

Acknowledgments

First and foremost, I would like to thank Dr. Ivelina Momcheva for welcoming me into her research group. I am deeply grateful for her unwavering positivity and support throughout my time in the group. She is not only an outstanding scientist and researcher but also a truly wonderful person—someone who was still providing me with corrections at 1:00 A.M. before submission.

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Finally, I want to thank my mom, whose calls I had to neglect at times but whose unwavering support I always felt, even from afar.

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established Helix at the University of Heidelberg. The code ran on this cluster.

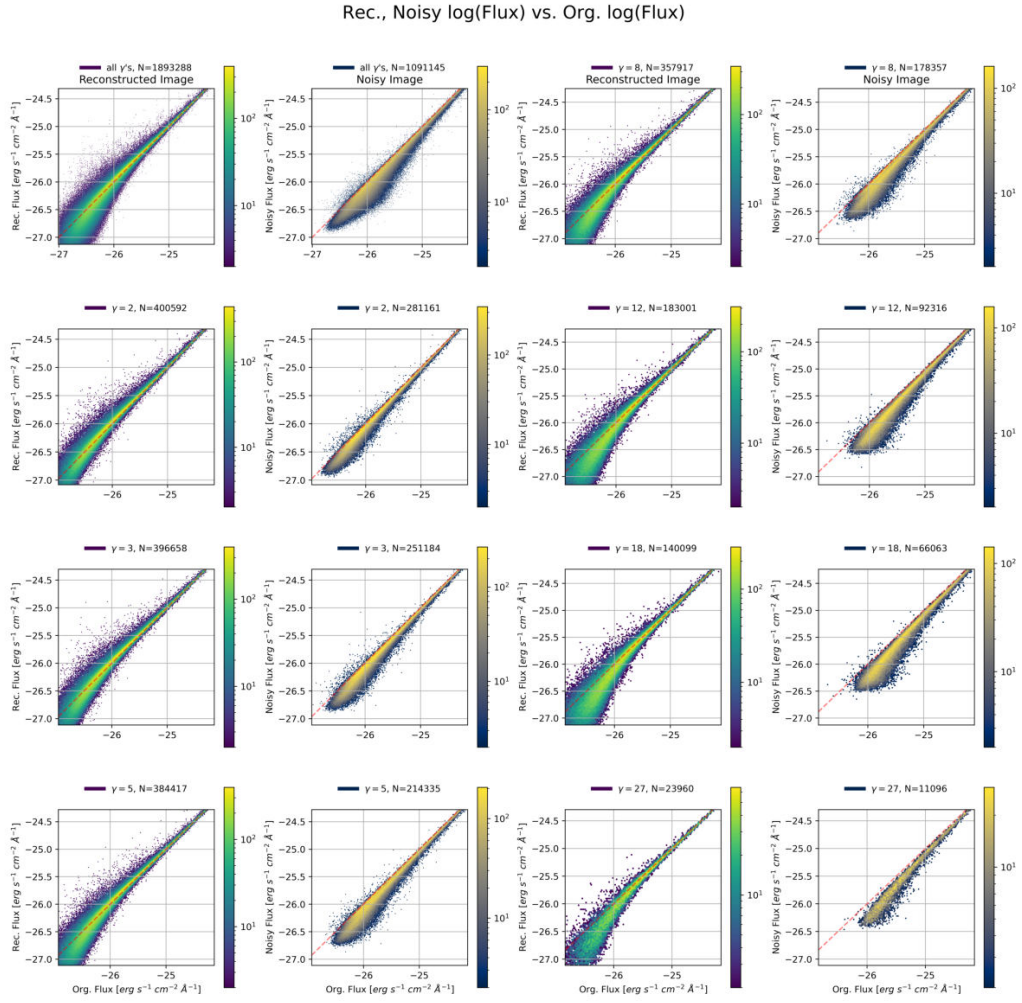


Figure 1: Figure shows the logarithm of flux for detections in reconstructed and noisy images compared to ground truth fluxes for the network trained on the larger dataset. Analogous to 4.5.

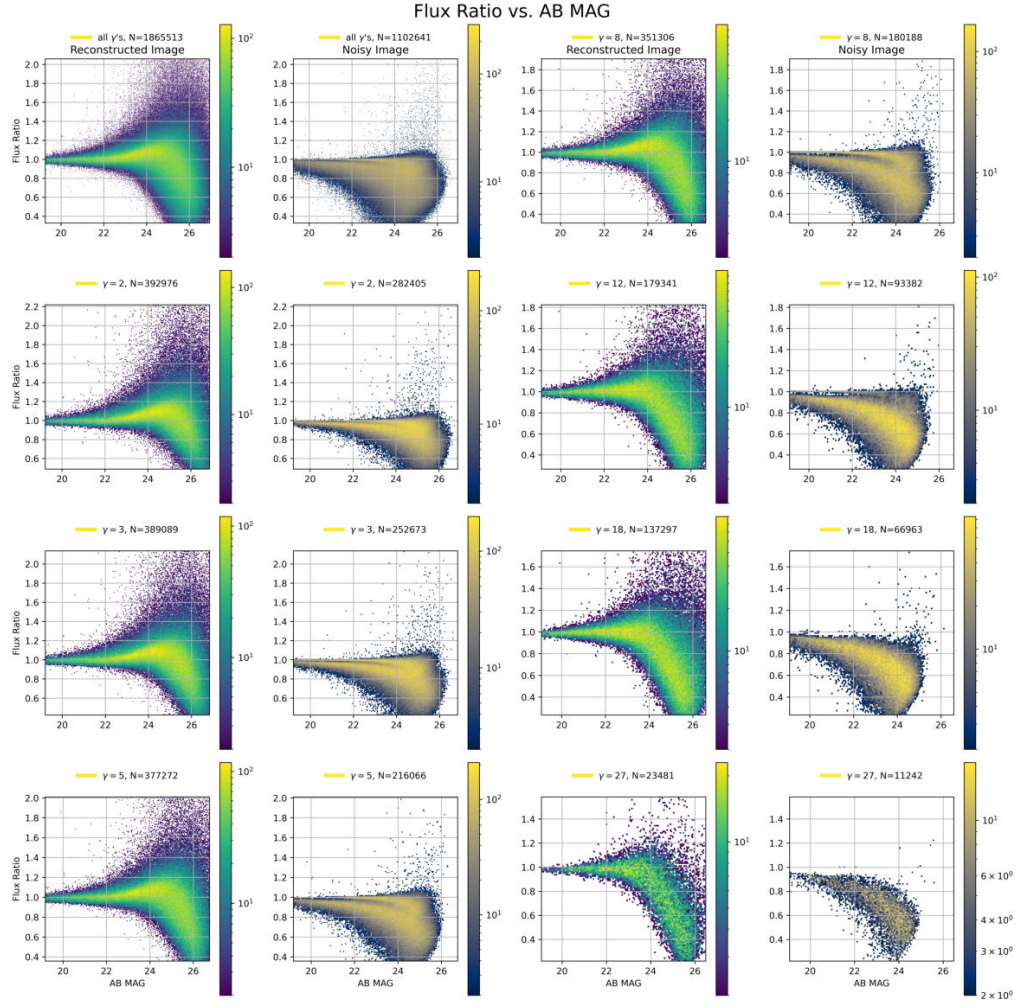


Figure 2: The flux ratios in reconstructed and noisy images show inverse distributions for the network trained on the larger dataset. Analogous to 4.6.

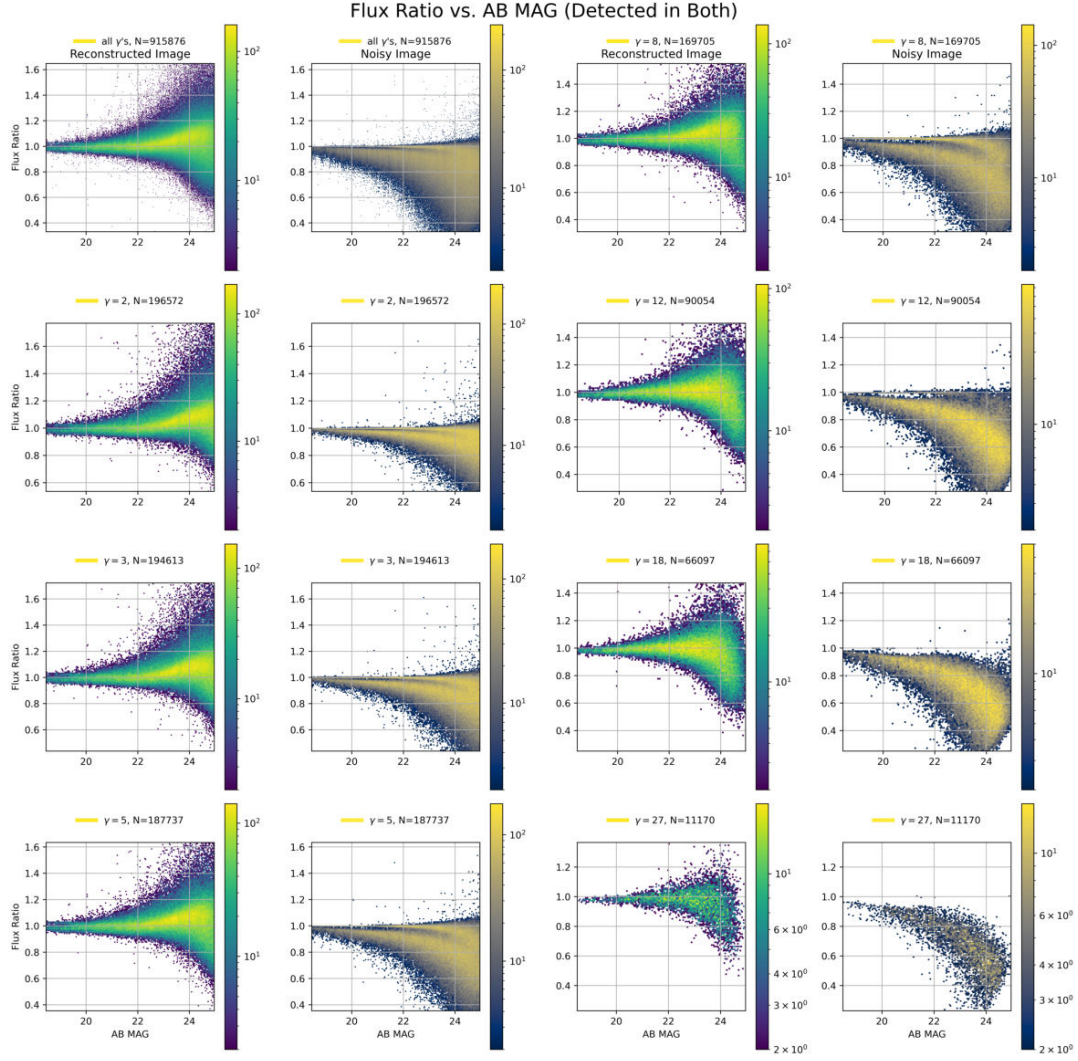


Figure 3: Figure shows the flux ratios for detections below 25 AB magnitude, where only objects detected in both reconstructed and original images are considered, analogous to 4.7.

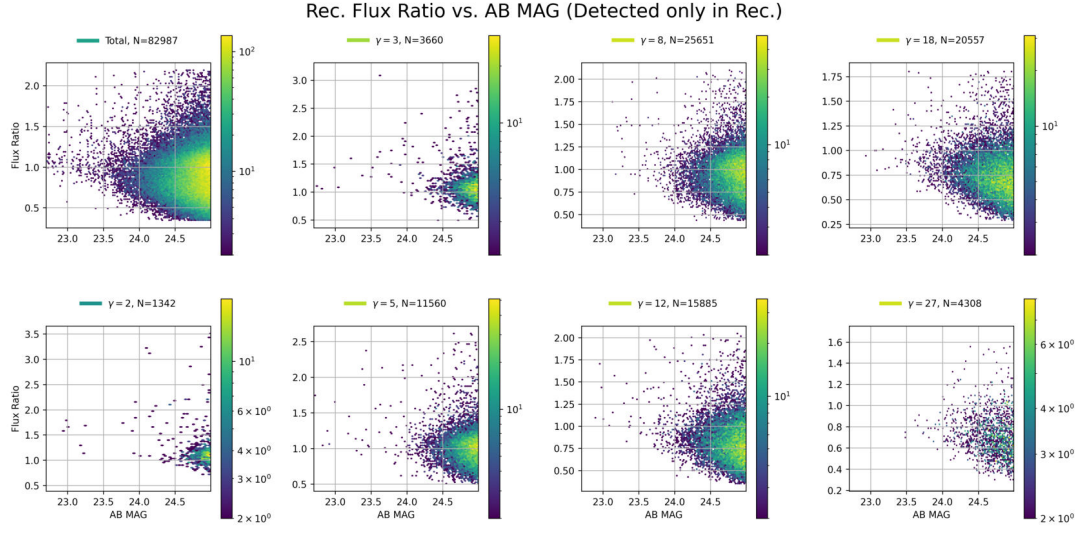


Figure 4: Figure shows the flux ratios of 83,000 detections with AB magnitudes below 25. Detections in the reconstructed images are mostly faint objects with AB magnitudes above 24, with flux ratios close to 1, though they can reach up to 2, in analogy to 4.8.

Rec., Noisy Kron Radius vs. Org. Kron Radius

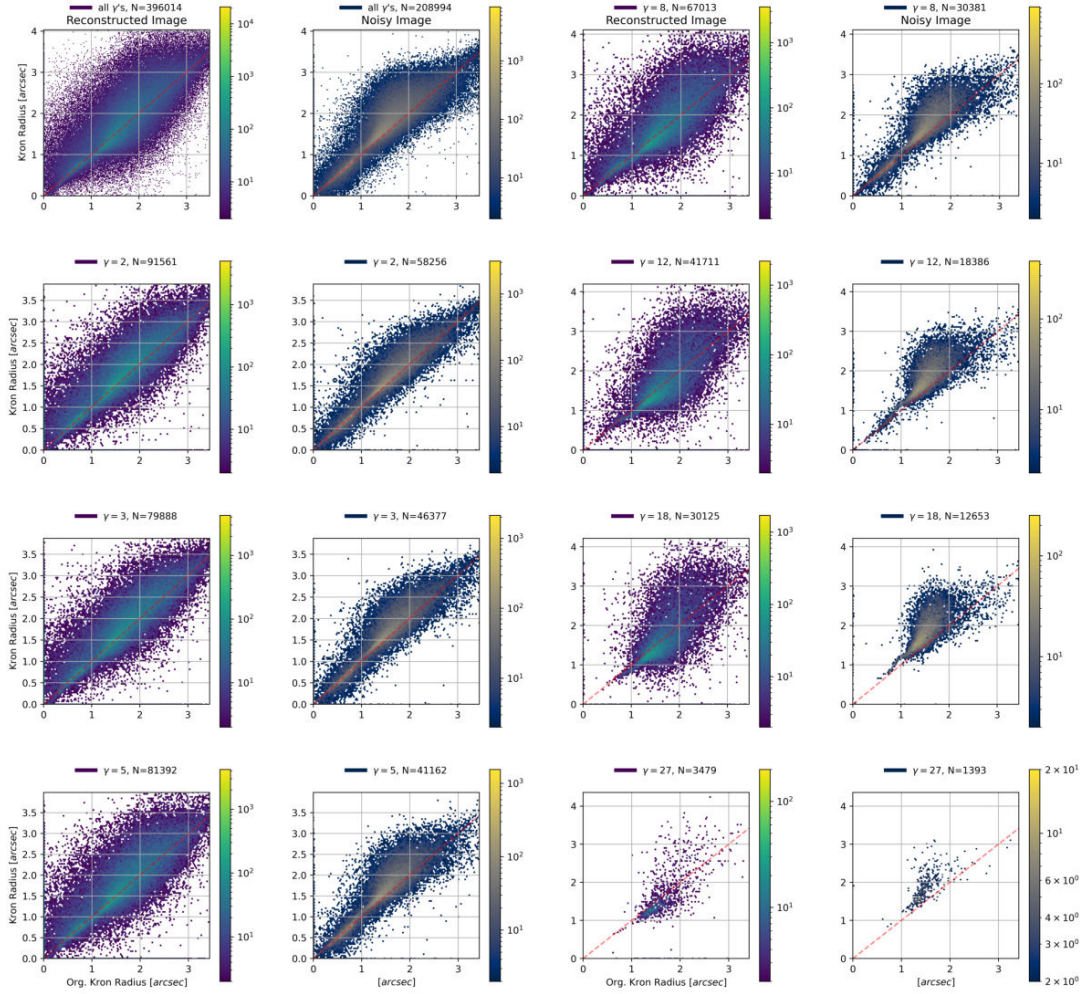


Figure 5: Comparison of the calculated Kron radius between reconstructed (1st and 3rd columns) and noisy (2nd and 4th columns) images along the y-axis, with ground truth images along the x-axis. The images are ordered by increasing exposure time ratio ($\gamma = \frac{t_{org}}{t_{rec}}$) from top to bottom. The red dashed line represents the one-to-one correspondence. In analogy to 4.9.

Rec., Noisy Δ Kron Radius vs. AB MAG

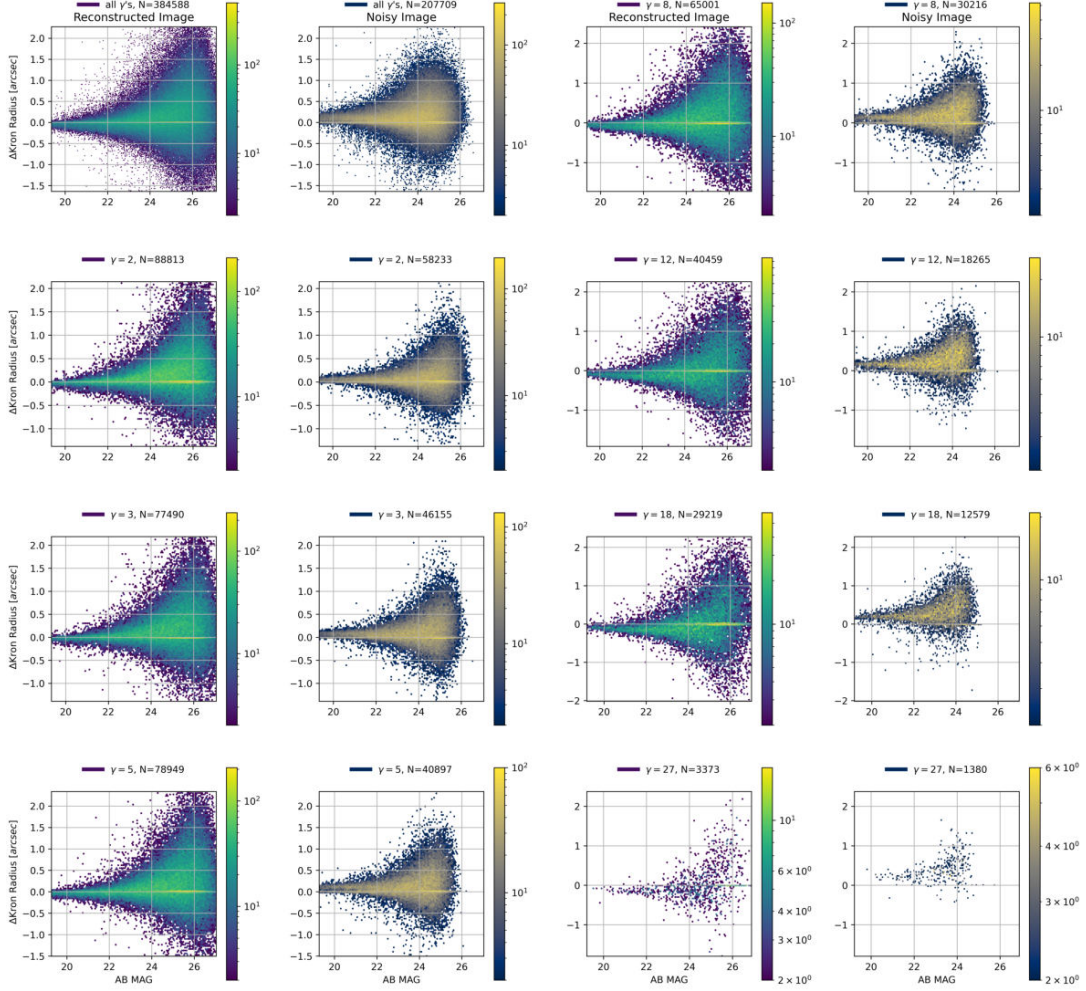


Figure 6: Difference in the Kron radius ($\Delta\text{Kron Radius} = \frac{\text{Kron Radius}_{\text{Rec,Noise}}}{\text{Kron Radius}_{\text{GT}}}$) between detections in the reconstructed (purple/green/yellow) and noisy (blue/yellow) images compared to ground truth, plotted against AB magnitude. The exposure time ratio increases from top to bottom for the network trained on the larger dataset in analogy to 4.10.

Rec., Noisy Kron Radius vs. Org. Kron Radius

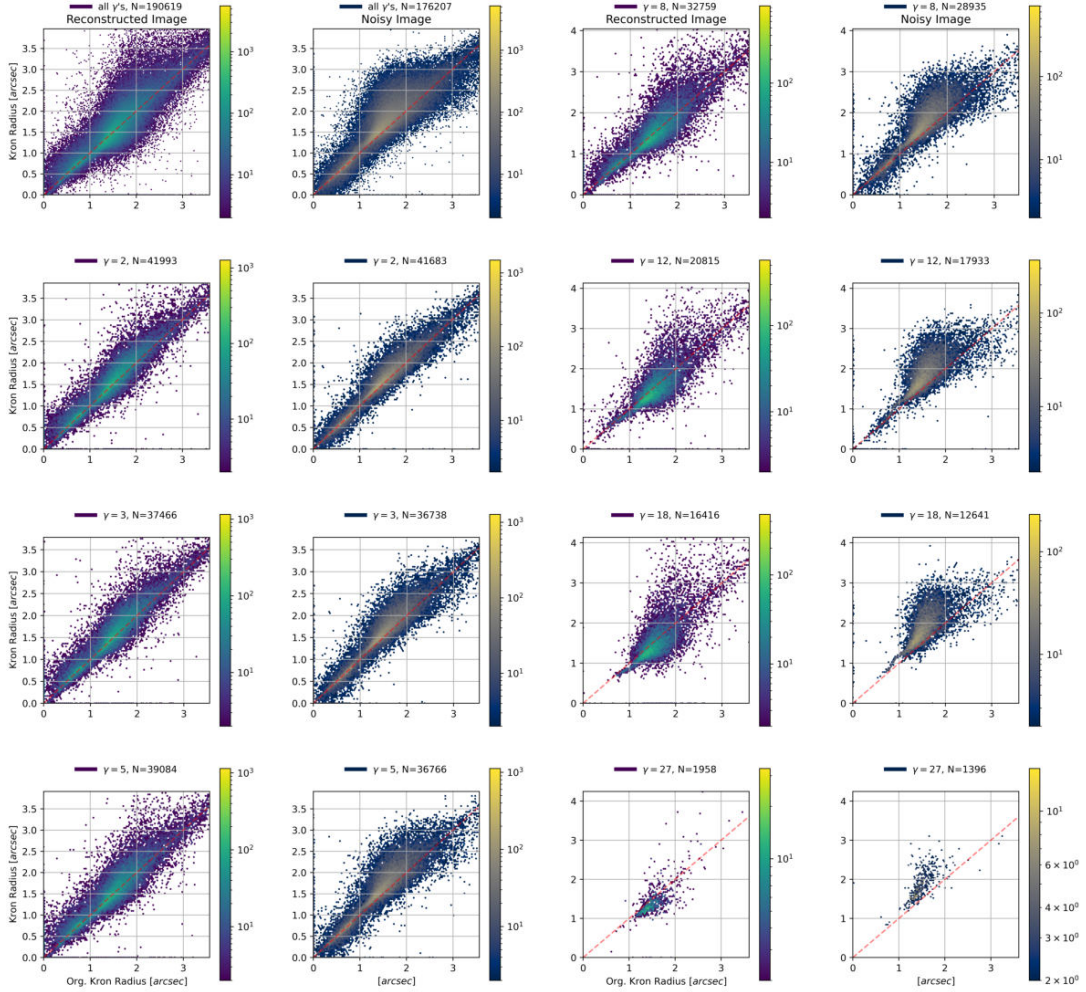


Figure 7: Similar to Figure 5, compares the Kron radius in reconstructed and noisy images to the original images, excluding detections above 25 AB magnitude for the network trained on the larger dataset. analogy with ??.

Rec., Noisy Δ Kron Radius vs. AB MAG (Detected in Both)

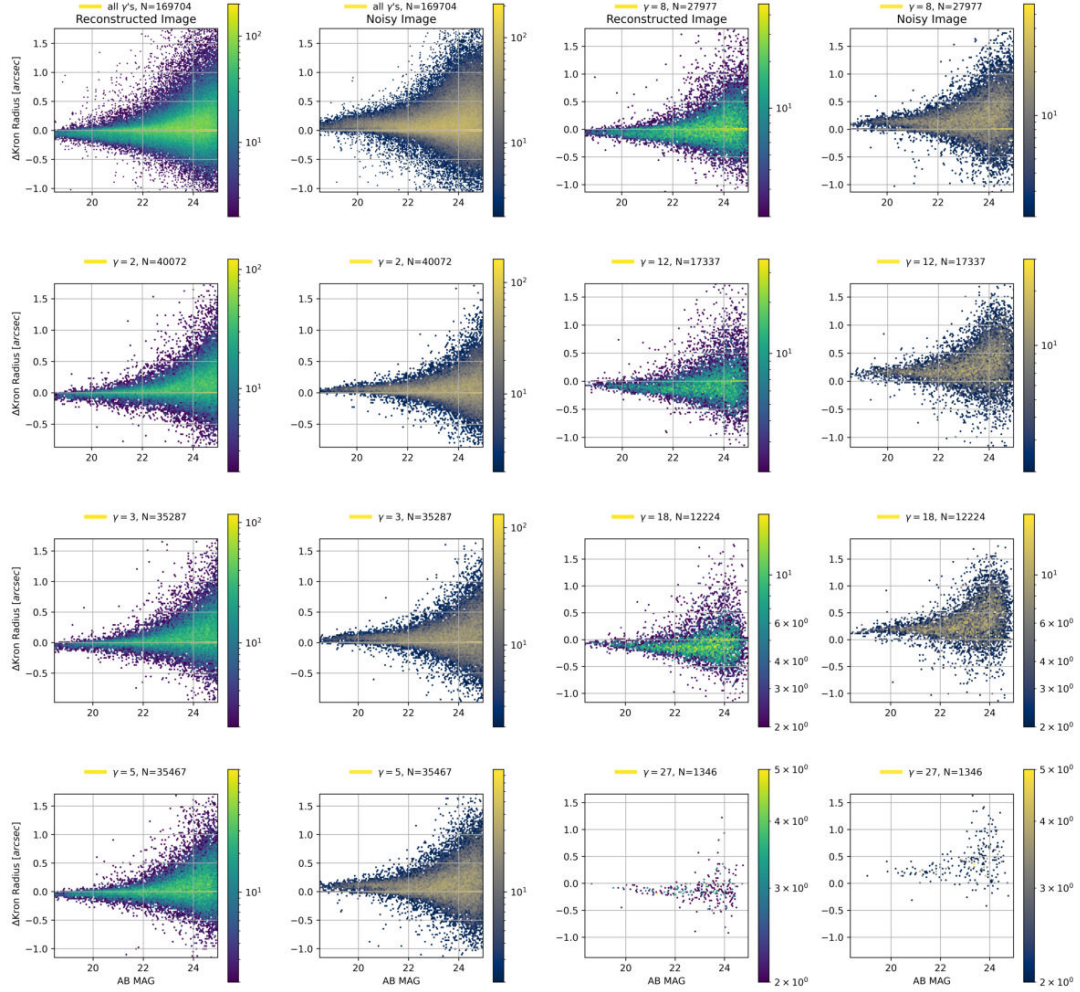


Figure 8: Figure shows the properties of Δ Kron for detections captured both in reconstructed. Analogous to 4.12.

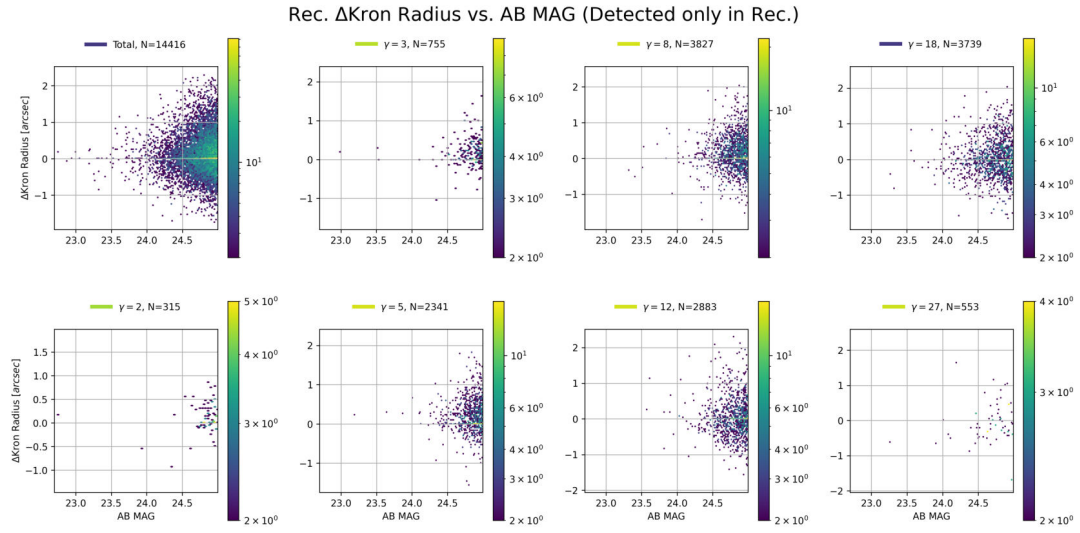


Figure 9: Figure shows detections below 25 AB magnitude, considering only objects found in reconstructed images, as there are no corresponding detections in noisy images. In analogy to 4.13.

Glossary

- DL: Deep Learning
- ML: Machine Learning
- API: Application Programming Interface
- HST: The Hubble Space Telescope
- WFC3 : Wide Field Camera 3
- IR: The Infrared Spectrum
- UVIS: The Ultraviolet and the Visible Spectrum
- NIR: Near-Infrared
- PN: Photon Noise
- RN: Readout Noise
- DN: Dark Current Noise
- CNN: Convolutional Neural Network
- RNN: Recurrent Neural Network
- MAE: Mean Absolute Error
- MSE: Mean Squared Error
- SGD: Stochastic gradient descent
- ADAM: Adaptive Moment Estimation
- SNR: Signal to Noise Ratio
- PSF: Point Spread Function
- SSIM: Structural Similarity Index
- pSNR: Peak Signal-to-Noise Ratio
- IoU: Intersection Over Union
- SEP: Source Extraction and Photometry

Eidesstattliche Erklärung

EIDESSTATTLICHE ERKLÄRUNG

Hiermit versichere ich, dass ich die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe. Alle Ausführungen, die anderen veröffentlichten oder nicht veröffentlichten Schriften wörtlich oder sinngemäß entnommen wurden, habe ich kenntlich gemacht.

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STATUTORY DECLARATION

I declare that I have authored this thesis independently, that I have not used other than the declared sources / resources and that I have explicitly marked all material which has been quoted either literally or by content from the used sources.

This paper was not previously presented to another examination board and has not been published.

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